

EMMA'S ATLAS OF NUMBERS AND SHAPES

- A NOT-SO-BRIEF GUIDE TO ALL THE MATH I KNOW -

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Part I

Foundations

Chapter 1

An Introduction to Mathematical Reasoning

Chapter 2

Sets

Almost all of modern mathematics can be put in the context of the study of *sets* and their *elements*. Generally, given a collection of objects $a, b, c, d, \dots$, they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \dots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, *Russel's paradox* asks:

"Let R be the set of all sets that don't contain themselves. Does R contain itself?"

If R doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If $R \in R$ is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of *axioms*, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics.

2.1 Infinity: $\emptyset$ and $\mathbb{N}$

2.2 Extensionality

To say " b is an element of M ", or equivalently " b contains M ", we simply write:

$$b \in M$$

One of the most basic axioms of set theory is the *Axiom of Extensionality*, which states:

Axiom 2.1: Axiom of Extensionality

Two sets A and B are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter. $\{1, 2\}$ and $\{2, 1\}$ are the same set.
2. The number of times an element appears within the same set doesn't matter. $\{1\}$, $\{1, 1\}$, and $\{1, 1, 1\}$ are the same set.

Definition 2.2: Subsets

We call a set A a *subset* of a set B if and only if every element of A is contained in B . We write " A is a subset of B " as $A \subseteq B$.

The demand that "every element of A is contained in B " can be stated more formally as " $x \in A$ implies $x \in B$ ". Importantly, this means that every set is a subset of itself - we always have $A \subseteq A$.

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets A and B are the same, then the easiest approach is almost always to show that any element of A must be contained in B , and to then show that any arbitrary element of B must be contained in A .

The subset relation is frequently written as $A \subset B$. However, other authors reserve $A \subset B$ to mean " A is a subset of B that isn't equal to B " - we call such a set a *proper subset* of B . I will therefore attempt to only use the symbols $\subseteq$ and $\subsetneq$, which clear up this ambiguity.

2.3 Specification

2.4 Pairing, Union, and Power Sets

2.5 Relations

2.6 Functions

2.7 Cartesian Products

Definition 2.3: General Products in Category Theory

Let C be a category. Let X_1 and X_2 be objects of C . Then the **product of X_1 and X_2** is an object $X_1 \times X_2$ equipped with two morphisms $\pi_1 : X \rightarrow X_1$, $\pi_2 : X \rightarrow X_2$ with the property that for every object Y and every pair of morphisms $f_1 : Y \rightarrow X_1$, $f_2 : Y \rightarrow X_2$, there exists a unique morphism $f : Y \rightarrow X_1 \times X_2$ such that:

1. $f_1 = \pi_1 \circ f$
2. $f_2 = \pi_2 \circ f$

We call this property the **universal property of product objects**.

In the case of sets, this definition leads us to the familiar cartesian products, where π_1 and π_2 are the projections to the first and second component of the product respectively. We can generalize the cartesian product to uncountably infinite products as follows:

Definition 2.4: Cartesian Product of sets

Let I be any set. Let $(X_i)_{i \in I}$ be a Family of nonempty sets indexed by I , i.e. there exists a function assigning to each element $i \in I$ an element of $(X_i)_{i \in I}$. Then the cartesian product of $(X_i)_{i \in I}$ is the unique set:

$$\prod_{i \in I} X_i := \left\{ x : I \rightarrow \bigcup_{i \in I} X_i \mid x_i := x(i) \in X_i \text{ for all } i \in I \right\}$$

Written more simply: Each element of $\prod_{i \in I} X_i$ is a function assigning to each $i \in I$ an element $x(i) \in X_i$, where we generally write the argument in the index ($x_i := x(i)$), to stay consistent with the established notation in the countable case.

This definition is simply a reinterpretation of the definition of finite cartesian products: It amounts to viewing, for example, the tuple $T := (x, y, z)$ as a map:

$$T : \{1, 2, 3\} \rightarrow \{x, y, z\}$$

such that $T(1) = x$, $T(2) = y$, and $T(3) = z$.

Definition 2.5: Projections

Let $J \subset I$. Then we define the projection of I to J as the map

$$\pi_J := \pi_J^I : \prod_{i \in I} X_i \rightarrow \prod_{j \in J} X_j$$

$$x \mapsto \left(x|_J : J \rightarrow \bigcup_{j \in J} X_j \right)$$

which restricts the domain of a given element x of the product to the subset J .

In particular, if J is a set consisting of a single element j , we have:

$$\pi_j := \pi_{\{j\}}^I : \prod_{i \in I} X_i \rightarrow X_j$$

$$x \mapsto x_j$$

2.8 Replacement

2.9 Regularity and Well-Foundedness

2.10 Choice

2.11 Cardinality

Part II

Foundations of Analysis

Chapter 1

The Real Numbers $\mathbb{R}$

1.1 Ordered Algebraic Structures

1.2 Dedekind Cuts

1.3 Uniqueness of $\mathbb{R}$

1.4 Cardinality of $\mathbb{R}$

Chapter 2

The Concept of Distance

2.1 Metric Spaces

A metric space is a set, alongside with a *distance function*, more commonly known as a *metric*, d which takes two points and outputs a distance between the points. Naturally, we want this "distance" to actually give us some usable structure on the set.

Now, what pieces of geometric intuition should we bake into our notion of distance? I would argue that the most fundamental use of "distance" is to be able to *compare distances between any two pairs of points*. Therefore, our metric should have as its output a *totally ordered set*.

We also want to be able to add distances together as a way of comparing them - logically, the distance between two points x and y plus the distance between y and z gives us an upper bound to the distance between x and z , since we can always take this path to get from x to z , but there might be a shorter path we haven't found yet. As a formula, this is the *triangle inequality*

$$d(x, z) \leq d(x, y) + d(y, z)$$

Finally, if we have an ordered group structure, we would really like for our order to be archimedean - the distance between two distinct points should be greater than zero, and it shouldn't be infinitesimal.

This leaves $\mathbb{R}_{\geq 0}$ as the obvious candidate for the domain of our distance function, since any archimedean ordered group is a subgroup of $\mathbb{R}_{\geq 0}$ anyways. This shouldn't be too much of a surprise, even without this logical background - I assume that when most people read "distance between two points", they intuitively already assume that these distances will be given by positive real numbers.

Definition 2.1: Metric Spaces

A *metric space* is a set M together with a function $d : M \times M \rightarrow \mathbb{R}_{\geq 0}$, such that:

1. d is *positive definite*, i.e. $d(x, y) = 0$ if and only if $x = y$.
2. d is *symmetric*, i.e. $d(x, y) = d(y, x)$
3. d fulfills the *triangle inequality*

$$d(x, z) \leq d(x, y) + d(y, z)$$

Theorem 2.2. Let $M^n := \prod_{i=1}^n M$ be a finite cartesian product of metric spaces. Then the *Euclidean distance function*

$$d_\infty(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

gives a metric on M^n .

Theorem 2.3. Let $M^n := \prod_{i=1}^n M$ be a finite cartesian product of metric spaces. Then the *Chebyshev distance function*

$$d_\infty(x, y) := \max_{i \in [1..n]} d(x_i - y_i)$$

gives a metric on M^n .

Definition 2.4: Balls

Let (M, d) be a metric space. Then the *open ball of radius r around a point x* is the set

$$B_r(x) := \{y \in M : d(x, y) < r\},$$

and the *closed ball of radius r around x* is the set

$$\overline{B}_r(x) := \{y \in M : d(x, y) \leq r\},$$

Definition 2.5: Open and Closed Sets

Let (M, d) be a metric space. Then we call a subset $U \subseteq M$ *open* (in M) if it contains an open ball (of any radius) around all of its points, i.e.

$$\forall x \in U :$$

$$\exists r \in \mathbb{R}_{\geq 0} :$$

$$B_r(x) \subseteq U.$$

We call U *closed* if its complement $M \setminus U$ is open.

Intuitively, a set is open if it contains "a little bit of wiggle room" around any point contained in it. This means that sets with a "clear border", such as our closed balls, aren't open. For example, the interval $[0, 1] \subset \mathbb{R}$ doesn't leave any wiggle room around 0 and 1.

Importantly, while it may seem unintuitive, there are in fact sets that are both closed and open at the same time. We call such sets *clopen*.

Exercise 2.6. *There are exactly two clopen subsets of $\mathbb{R}$. What are they?*

Definition 2.7. Let (M, d) be a metric space. We call a subset $N \subseteq M$ *dense in M* if every nonempty open subset of M contains an element of N .

Lemma 2.8. Every nonempty open subset of $\mathbb{R}$ is uncountable.

Proof. 1. Since every nonempty open subset of $\mathbb{R}$ contains an open interval (a, b) , it suffices to show that open intervals are uncountable.

2. The map

$$\begin{aligned}\varphi_1 : (a, b) &\rightarrow \left(-\frac{b-a}{2}, \frac{b-a}{2}\right) \\ x &\mapsto x - \frac{b+a}{2}\end{aligned}$$

is a bijection.

3. The map

$$\begin{aligned}\varphi_2 : \left(-\frac{b-a}{2}, \frac{b-a}{2}\right) &\rightarrow (-1, 1) \\ x &\mapsto \frac{2x}{b-a}\end{aligned}$$

is also a bijection.

4. Finally, the map

$$\begin{aligned}\varphi_3 : (-1, 1) &\rightarrow \mathbb{R} \\ x &\mapsto \frac{x}{1-|x|}\end{aligned}$$

is a bijection (the inverse is the map we get by replacing the $-$ with a $+$).

5. Therefore, $\varphi_3 \circ \varphi_2 \circ \varphi_1$ is a bijection $(a, b) \cong \mathbb{R}$, so any open interval is uncountable, so any open subset of $\mathbb{R}$ is uncountable. $\square$

Theorem 2.9

The set of rational numbers $\mathbb{Q}$ is dense in $\mathbb{R}$. The set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$.

Proof. 1. Since any nonempty open subset $U \subseteq \mathbb{R}$ contains an open interval (a, b) , it suffices to show that any open interval contains a rational number. We assume without loss of generality that $0 \leq$

$a < b$, since $a < 0 < b$ already implies $0 \in (a, b)$, while $a < b \leq 0$ implies that (a, b) contains a rational number if and only if $(-b, -a)$ contains a rational number.

Now, since the ordering on the reals is archimedean, we can find a natural number $n \in \mathbb{N}$ such that $\frac{1}{n} < b - a$. Now, the set

$$L = \left\{ k \in \mathbb{N} \mid \frac{k}{n} > a \right\}$$

is nonempty - if $a = 0$, then $1 \in M$, and if $a \neq 0$, then the archimedean property tells us that there exists a $k \in \mathbb{N}$ such that

$$\frac{1}{k} < \frac{1}{na},$$

which implies $\frac{k}{n} > a$. Since M is a nonempty subset of $\mathbb{N}$, there must be a smallest element m . Since m is minimal, we have $m - 1 \notin M$, which implies

$$\frac{m-1}{n} \leq a,$$

which in turn implies

$$\frac{m}{n} = \frac{m-1}{n} + \frac{1}{n} < a + (b-a) = b$$

Therefore, $\frac{m}{n}$ is a rational number contained in (a, b) , so $\mathbb{Q}$ is dense in $\mathbb{R}$.

2. Since any open subset $U \subseteq \mathbb{R}$ is uncountable, while the rationals are countable, U must contain an uncountable number of irrationals. Therefore, $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

□

Corollary 2.10

The set of rational numbers $\mathbb{Q}$ is neither closed nor open in $\mathbb{R}$.

Proof. Since both sets are dense in $\mathbb{R}$, every open ball containing a rational number must also contain an irrational number, and vice versa. □

2.2 Topologies

Topological spaces generalize metric spaces. By inheriting the concept of open and closed sets from metric spaces, they are able to still capture notions of closeness without putting a numerical value on exactly how close any given pair of points are. The notion of a topology is based on the following insight:

Exercise 2.11. Let (M, d) be a metric space. Then:

1. *The empty set is open in M .*
2. *Any union of sets that are open in M is open in M .*
3. *Any finite intersection of sets that are open in M is open in M .*
4. *M is open in M .*

We take this idea and make it a definition:

Definition 2.12

Let X be a set. Then a **topology on X** is a set $\mathcal{O}$ of subsets of X , called **open sets**, such that:

1. The empty set is open.
2. Any union of open sets is open.
3. Any finite intersection of open sets is open.
4. X is open.

Most of the topological spaces we care about in practice are already metric spaces, but many proofs flow more elegantly when phrased in the generalized language of topological spaces, and there *are* topological spaces that aren't metric spaces.

Theorem 2.13: Extremal Topologies

Let X be an arbitrary set. Then the following two sets of subsets are always topologies on X :

1. The **indiscrete topology**, where the only open sets are X and the empty set.
2. The **discrete topology**, where every subset is open.

Proof Sketch. Both topologies fulfill all requirements trivially. :) "□"

Definition 2.14. We say that a metric d on a space M **generates** a given topology $\mathcal{O}$ on M if they give the same open sets, i.e. the open sets in $\mathcal{O}$ are exactly the open sets in (M, d) .

Exercise 2.15.

1. Show that the discrete metric generates the discrete topology.
2. Show that there is no metric generating the indiscrete topology.

Two different metrics can generate the same topology:

Exercise 2.16. Show that the euclidean metric also generates the discrete topology on $\mathbb{N}$.

Exercise 2.17. Let d be a metric on a set M . Let $c \in \mathbb{R}_{>0}$.

1. Show that

$$d_c(x, y) = c \cdot d(x, y)$$

is a metric on M .

2. Show that d and d_c generate the same topologies on M .

This fact can be seen as a formalization of the intuition that changing the specific unit we choose to measure space with doesn't change its underlying structure.

Definition 2.18: Interior, Exterior and Boundary

Let X be a topological space. Let $A \subseteq X$. Then:

1. The **interior** $\text{int } A$ of A is the set of all points that have an open neighbourhood entirely contained in A .
2. The **exterior** $\text{ext } A$ of A is the interior of the complement $X \setminus A$, i.e. the set of all points that have an open neighbourhood entirely contained in $X \setminus A$.
3. The **boundary** ∂A of A is the set of all points $x \in X$ lying neither in $\text{int } A$, nor in $\text{ext } A$.

The interior of A is occasionally written as $\overset{\circ}{A}$. Intuitively, the interior is the largest open subset of A , and the exterior is the largest open set not containing any point of A .

Exercise 2.19. Show that:

1. The interior of any set is open.
2. The exterior of any set is open.
3. The interior of A is the union of all open subsets of A .
4. The exterior of A is the union of all open subsets of $X \setminus A$.
5. The boundary of A is the set of all points $x \in X$ such that every neighbourhood of x contains both points in A and points in $X \setminus A$.

Definition 2.20: Closure

The closure of A is the set $\overline{A} := A \cup \partial A$ containing A and all of its boundary points.

Exercise 2.21.

1. Show that $\overline{A}$ is the set of all points $x \in X$ such that every neighbourhood of x contains points in A .
2. Show that $\overline{A}$ is closed.
3. Show $\overline{A}$ is the intersection of all closed sets containing A .

Theorem 2.22. Let $B \subseteq \mathbb{R}$ be a bounded subset of $\mathbb{R}$. Then $\inf B$ and $\sup B$ are contained in $\overline{B}$.

Proof. Since B is bounded, completeness of the real numbers tells us that $\inf B$ and $\sup B$ are finite real numbers. If $\inf B \notin B$, then there would exist an open neighbourhood $U_{\inf}$ of $\inf B$ not containing any points of B . Since any neighbourhood of a real number x must contain numbers greater than x , this would give us a lower bound of B greater than $\inf B$, which contradicts the definition of $\inf B$. Similarly, if $\sup B \notin \overline{B}$, we could get a lesser upper bound of B , which is again a contradiction. $\square$

2.3 Convergence

Theorem 2.23: Detecting openness and closedness using sequences

Let M be a metric space. Let $A \subseteq M$. Let $x \in M$. Then:

1. $x \in \text{int } A$ if and only if every sequence in M converging to x is eventually in A .
2. $x \in \overline{A}$ if and only if there exists a sequence in A converging to x .

Proof. 1. If $x \in \text{int } A$, then there exists an $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq A$. If a sequence converges to x , it must thus be eventually in $B_\varepsilon(x)$, and thus eventually in A .

Vice versa, if $x \notin \text{int } A$, then every open ball $B_n := B_{\frac{1}{n}}(x)$ around A contains at least one point not in A , and so we can construct a sequence converging to x that isn't eventually in A by picking these points.

2. If there exists sequence of points in A converging to x , then every neighbourhood of x must contain at least one point in A , so $x \in \overline{A}$.

Vice versa, if $x \in \overline{A}$, then every open ball around x must contain a point from A , so we can construct a sequence in A converging to x by taking the sequence $B_n := B_{\frac{1}{n}}(x)$ of open balls and picking a point $x_n \in A \cap B_n$ for each $n \in \mathbb{N}$.

□

Corollary 2.24.

1. A is closed in X if and only if it contains every limit of every convergent sequence of points in A .
2. A is open in X if and only if every sequence in X converging to a point of A is eventually in A .

2.4 Continuity

Definition 2.25. Let $f : M \rightarrow N$ be a function between metric spaces. Then we call f *uniformly continuous* if it is continuous and fulfills the additional requirement that, given an ε , we can pick the same δ for all points $x \in M$, i.e:

$$\forall \varepsilon \in \mathbb{R}_{>0} :$$

$$\exists \delta \in \mathbb{R}_{>0} :$$

$$\forall x, y \in M :$$

$$d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon$$

2.5 Compactness

Exercise 2.26. Any closed subset of a compact set is compact.

Exercise 2.27. A compact subset of a metric space is always closed.

Theorem 2.28: Compactness in Metric Spaces

1. A subset of a metric space is compact if and only if it is sequentially compact.
2. A compact subset of a metric space is closed and bounded.

Proof. Let $K \subseteq M$ be compact. Since M is a metric space, K is closed. If we cover K using open balls, that cover must have a finite subcover. Thus, K must be contained in a finite union of open balls, which means K must be bounded. $\square$

Theorem 2.29

Let $f : M \rightarrow N$ be a continuous function between metric spaces. Let $K \subseteq M$ be compact. Then $f(K)$ is compact.

Proof. Let $K \subseteq M$ be compact. Let $\mathcal{U}$ be an open cover of $f(K)$. Since f is continuous, $f^{-1}[\mathcal{U}] := \{f^{-1}[U] \mid U \in \mathcal{U}\}$ is an open cover of K , which must have a finite subcover $\mathcal{V} \subseteq f^{-1}[\mathcal{U}]$. Since $f : K \rightarrow f(K)$ is surjective by definition, $f[\mathcal{V}]$ must be a finite cover of $f(K)$. $\square$

Corollary 2.30: Maximum-Minimum-Theorem

Let $f : K \rightarrow \mathbb{R}$ be a function from a compact subset of a metric space into the real numbers. Then f attains a maximum and a minimum.

Proof. By the previous theorem, $f(K)$ is a compact subset of $\mathbb{R}$, i.e. closed and bounded by Theorem 2.28. But then Theorem 2.22 tells us that $\inf f(K)$ and $\sup f(K)$ are already contained in $\overline{f(K)} = f(K)$. $\square$

Lemma 2.31. Let M, N be metric spaces. Let $K \subseteq M$ be compact. Let $f : K \rightarrow N$ be continuous. Then the preimage of any compact subset of N is a compact subset of K .

Proof. Since N is a metric space, compact subsets are closed. Since f is continuous, the preimage of a closed subset is closed. Since K is compact, any closed subset of K is compact. $\square$

Theorem 2.32: Countable Metric Tychonoffs Theorem

Let M_1, M_2 be metric spaces. Let $K_1 \subseteq M_1, K_2 \subseteq M_2$ be compact. Then $K_1 \times K_2 \subseteq M_1 \times M_2$ is compact.

Proof. Using whichever bijection $\mathbb{N}^2 \cong \mathbb{N}$ the reader prefers, we can unroll a sequence s_n in $K_1 \times K_2$ into a sequence x_n in K_1 and a sequence y_n in K_2 , such that the elements of s_n are in one-to-one correspondence with pairs (x_i, y_j) of elements of x_n and y_n . Since K_1 and K_2 are compact, x_n and y_n have convergent subsequences. Using our bijection, these can be reassembled into a subsequence of s_n , and since both components of this subsequence converge, the subsequence itself converges. Thus, every sequence in $K_1 \times K_2$ has a convergent subsequence. Since sequential compactness and compactness are equivalent in metric spaces by Theorem 2.28, $K_1 \times K_2$ is compact. $\square$

Theorem 2.33

Let M, N be metric spaces. Let $K \subseteq M$ be compact. Then every continuous function $f : K \rightarrow N$ is uniformly continuous.

Proof. Assume that $f : K \rightarrow N$ isn't uniformly continuous. Then there exists an $\varepsilon \in \mathbb{R}_{>0}$ such that for every $n \in \mathbb{N}$, we can find points $x_n, y_n \in K$ such that $d(x_n, y_n) < 1/n$, but $d(f(x_n) - f(y_n)) \geq \varepsilon$.

Since K is compact, it is sequentially compact by Theorem 2.28, so there exists a subsequence $(x_k) \subset (x_n)$ converging to a point p . Since $d(x_n, y_n) < 1/n$, there also exists a subsequence $x_k \subset y_n$ converging to p . If f was continuous, we would have

$$\lim_{k \rightarrow \infty} f(x_k) = \lim_{k \rightarrow \infty} f(y_k),$$

contradicting the fact that we have $d(f(x_k) - f(y_k)) \geq \varepsilon$ for all $k \in \mathbb{N}$. $\square$

2.6 Connectedness

Definition 2.34

A metric space is called **connected** if it can't be written as the disjoint union of two nonempty open subsets.

Exercise 2.35. 1. Show that $\mathbb{N}$ is not connected.

2. Show that $\mathbb{Q}$ is not connected.

Exercise 2.36. Show that a metric space M is connected if and only if its only clopen subsets are M and $\emptyset$.

Theorem 2.37: Connected Subsets of $\mathbb{R}$

The connected subsets of $\mathbb{R}$ are the intervals $I \subseteq \mathbb{R}$.

Proof. 1. Let $I \subseteq \mathbb{R}$ be an interval. Assume we have a decomposition

$$I = I_1 \cup I_2$$

of I into disjoint nonempty open subsets. Without loss of generality, assume that there exists $x_1 \in I_1$ and $x_2 \in I_2$ such that $x_1 < x_2$ (otherwise swap the two subsets). Now, consider the set

$$S = \{x \in I_1 \mid x \leq x_2\}$$

S is nonempty, since $x_1 \in S$, and S has x_2 as an upper bound. Now, let $s = \sup S$. We have $x_1 \leq s \leq x_2$, so $s \in I$. By Theorem 2.22, we have $s \in \bar{S} \subseteq \bar{I}_1$.

Since we have $I_1 = I \setminus I_2$, we know that I_1 is closed in I , giving us $s \in \bar{I}_1 = I_1$.

Since I_1 is by definition also open in I , and $s \leq x_2$ cannot be an upper bound of I , there exists $\varepsilon \in \mathbb{R}_{>0}$ such that $s + \varepsilon \in I_1$, which contradicts $s = \sup I_1$. Therefore, all intervals must be connected.

2. Let $C \subseteq \mathbb{R}$ be connected. If C is empty or only contains one point, it vacuously qualifies as an interval (x, x) or $[x, x]$. Now, assume that C contains at least two points, but isn't an interval. Then there exist $c_1 \neq c_2 \in C$ and $c \in [c_1, c_2]$ such that $c \notin C$. But then

$$C_1 = (-\infty, c) \cap C$$

$$C_2 = (c, \infty) \cap C$$

is a decomposition of C into disjoint nonempty open subsets.

□

Theorem 2.38

Let $f : M \rightarrow N$ be a continuous map between metric spaces. Let M be connected. Then $f(M)$ is connected.

Proof. Assume $f(M)$ wasn't connected. Then we would have some decomposition $f(M) = U \cup V$ of $f(M)$ into disjoint open sets. Since f is continuous, $f^{-1}(U)$ and $f^{-1}(V)$ are open. But then $M = f^{-1}(U) \cup f^{-1}(V)$ can't be connected. $\square$

Corollary 2.39: Intermediate Value Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous map from a closed real interval into the real numbers. Let y be some number between $f(a)$ and $f(b)$ ¹. Then there exists $x \in [a, b]$ such that $f(x) = y$.

¹ i.e. $f(a) \leq y \leq f(b)$ or $f(b) \leq y \leq f(a)$

Proof. Since f is continuous, $f([a, b])$ is a connected subset of $\mathbb{R}$, i.e. an interval. Since $f([a, b])$ contains $f(a)$ and $f(b)$, it must therefore also contain all numbers in between the two. $\square$

Corollary 2.40: Bolzano's Theorem

If a continuous function $f : M \rightarrow \mathbb{R}$ has a point $x \in M$ with $f(x) < 0$ and a point $y \in M$ with $f(y) > 0$, there exists a root $z \in M : f(z) = 0$ between x and y .

- Exercise 2.41.**
1. Assuming that temperature varies continuously, show that there are always two points at a given latitude on earth at which the temperature is the same.
 2. Assuming there are two points on earth that are a given distance apart from each other, are there always two points on earth that are that distance apart from each other and have the same temperature?

2.7 Function Spaces

Since we know that a sequence in a finite cartesian product converges if and only if all components converge, it seems natural to generalize this notion to arbitrary, potentially infinite cartesian products, i.e. arbitrary functions:

Definition 2.42: Pointwise Convergence

We say that a sequence of functions $f_n : M \rightarrow N$ converges *pointwise* to a given function $f : M \rightarrow N$ if and only if all components $f_n(x)$ converge in the usual sense, i.e:

$$\begin{aligned} \forall x \in M : \\ \forall \varepsilon \in \mathbb{R}_{>0} : \\ \exists N \in \mathbb{N} : \\ \forall n \geq N : \\ d(f_n(x) - f(x)) < \varepsilon \end{aligned}$$

We can't put a metric on N^M in which this form of converges matches regular metric convergence, but the intuition that this is the most natural form of convergence in N^M will be vindicated once we study abstract product spaces in point-set topology, at which point we confirm that in the natural product topology, an infinite cartesian product still converges if and only if every component converges individually.

However, there *is* a natural metric to put on N^M , which leads to a stronger form of convergence:

Definition 2.43. An *extended metric space* is a metric space where d can also take on the value ∞ .

Theorem 2.44: Supremum Metric

The set N^M of functions from a metric space M to a metric space N can be metrized by the extended metric

$$d_\infty(f, g) := \sup_{x \in M} |f(x) - g(x)|.$$

Additionally, this function is a regular metric on the set $B(M, N) \subset N^M$ of bounded functions $M \rightarrow N$.

Note that is simply a direct generalization of the Chebyshev metric 2.3 to infinite cartesian products!

Theorem 2.45: Uniform Convergence

In (N^M, d_∞) , a sequence of functions f_n converges to a given function f if and only if

$$\begin{aligned} \forall \varepsilon \in \mathbb{R}_{>0} : \\ \exists N \in \mathbb{N} : \\ \forall n \geq N : \\ \forall x \in M : \\ d(f_n(x) - f(x)) < \varepsilon \end{aligned}$$

we call this form of convergence of a sequence of functions *uniform convergence*.

Theorem 2.46: Uniform Convergence of Continuous Functions

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of continuous functions $f_n : M \rightarrow N$ converging uniformly to a function $f : M \rightarrow N$. Then f is continuous.

Proof. Choose some $\varepsilon \in \mathbb{R}_{>0}$. Since $(f_n)_{n \in \mathbb{N}}$ converges to f uniformly, there exists an $N \in \mathbb{N}$ such that

$$\begin{aligned} \forall x \in M : \\ n \geq N \implies d(f(x), f_n(x)) < \varepsilon/3 \end{aligned}$$

Let $x_0 \in M$ be arbitrary. Since f_N is continuous, there exists a $\delta \in \mathbb{R}_{>0}$ such that

$$\begin{aligned} \forall x \in M : \\ d(x, x_0) < \delta \implies d(f_N(x), f_N(x_0)) < \varepsilon/3 \end{aligned}$$

Applying the triangle inequality twice gives

$$\begin{aligned} d(f(x), f(x_0)) \\ \leq d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) \end{aligned}$$

to which we can apply the first inequality, then the second, and then the first again, to get:

$$\begin{aligned} d(f(x), f(x_0)) \\ \leq d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) \\ < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon \end{aligned}$$

for all $x \in M$ with $d(x, x_0) < \delta$. □

Theorem 2.47. Let $U \subseteq (X, d)$ be open and path-connected. Then there exists a piecewise linear path $\gamma_n : [0, 1] \rightarrow U$ between any two $p, q \in U$.

Proof Sketch. 1. Since U is path connected, there exists a curve $\gamma : [0, 1] \rightarrow U$ connecting p and q .

2. We can approximate γ via a piecewise linear curve γ_n by setting

$$\gamma_n\left(\frac{x}{n}\right) := \gamma\left(\frac{x}{n}\right)$$

3. Since γ is continuous and defined on a compact set, it is uniformly continuous by Theorem 2.5. From this, it can be shown that the γ_n converge to γ uniformly.

4. Since $\gamma([0, 1]) \subseteq U$ is compact and $X \setminus U$ is closed and disjoint from U , we have $\varepsilon := d(\gamma([0, 1]), X \setminus U) > 0$.

5. Therefore, uniform convergence guarantees there is an n such that $d(\gamma_n([0, 1]), \gamma([0, 1])) < \varepsilon$, which implies $\gamma_n([0, 1]) \subseteq U$.

”□”

2.8 Banach's Fixed Point Theorem

Chapter 3

Infinite Series

3.1 Achilles and the Turtle

3.2 Convergence Tests

3.3 Euler's Number e

3.4 The Exponential Function

Chapter 4

Euclidean Geometry

4.1 The Complex Numbers $\mathbb{C}$

4.2 Circles, Angles and Trigonometry

Part III

Multivariable Calculus

Chapter 1

Differentiation in $\mathbb{R}^n$

1.1 Differentials and Partial Derivatives

Definition 1.1: Fréchet Derivative

Let V, W be normed vector spaces. Let U be an open subset of V . Then $f : U \rightarrow W$ is called *Fréchet differentiable* or just *differentiable* at $p \in U$ if there exists a continuous linear map $Df(p) : V \rightarrow W$ such that

$$\lim_{h \rightarrow 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|_W}{\|h\|_V} = 0.$$

We call $Df(p)$ the *differential of f at p* . We call f *differentiable* if it is differentiable at every point $p \in U$. We call f *continuously differentiable* if the map $Df : p \mapsto Df(p)$ is continuous.

Recall that linear maps are continuous if and only if they are bounded, which is always the case if V is finite-dimensional.

Exercise 1.2. Verify that if we identify scalars $c \in \mathbb{R}$ with linear maps $x \mapsto c \cdot x$, this definition agrees with the usual definition for $f : \mathbb{R} \rightarrow \mathbb{R}$.

Theorem 1.3. If $f : U \rightarrow W$ is differentiable, it is continuous.

Proof. If f is differentiable, we have

$$\lim_{\|h\|_V \rightarrow 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|_W}{\|h\|_V} = 0.$$

This can only be the case if

$$\lim_{\|h\|_V \rightarrow 0} \|f(p+h) - f(p) - Df(p)(h)\|_W = 0,$$

i.e. if

$$\lim_{\|h\|_V \rightarrow 0} f(p+h) = \lim_{\|h\|_V \rightarrow 0} f(p) + Df(p)(h).$$

Since $Df(p)$ is continuous with $Df(0) = 0$, this simplifies to

$$\lim_{\|h\|_V \rightarrow 0} f(p+h) = f(p),$$

which is the definition of sequential continuity. $\square$

Definition 1.4: Directional and Partial Derivates

Let V and W be normed vector spaces. Let $v \in V$. Let $U \subseteq V$ be open and $f : U \rightarrow W$. Then we call the limit

$$\frac{\partial}{\partial v} f(p) = \left. \frac{d}{dh} f(p + hv) \right|_{h=0} = \lim_{h \rightarrow 0} \frac{f(p + hv) - f(p)}{h}$$

the **directional derivative** of f at p in direction v . If $(x_0, \dots, x_n)$ is a basis of V , we call the directional derivatives

$$\partial_i f(p) := \frac{\partial}{\partial x_i} f(p)$$

the **partial derivatives of f at p** . If all partial derivatives exist, we call f **partially differentiable**. If all maps $\partial_i f : p \mapsto \partial_i f(p)$ are continuous, we call f **continuously partially differentiable**.

Partial derivatives aren't actually any more complicated than standard one-dimensional derivatives - you can calculate partial derivatives by simply fixing all coordinates except one and then taking a regular derivative in the remaining coordinate. For example, if we have a function

$$\begin{aligned} f : \mathbb{R}^3 &\rightarrow \mathbb{R} \\ (x, y, z) &\mapsto f(x, y, z), \end{aligned}$$

we can define a function $f|_1 : \mathbb{R} \rightarrow \mathbb{R}$ by restriction to the first coordinate:

$$f|_1(x) := f(x, y, z).$$

Using this function, we can simplify the partial derivative:

$$\begin{aligned} \partial_1 f(x, y, z) &= \lim_{h \rightarrow 0} \frac{f((x, y, z) + he_1) - f(x, y, z)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x + h, y, z) - f(x, y, z)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f|_1(x + h) - f|_1(x)}{h} \\ &= \frac{d}{dx} f|_1(x) \end{aligned}$$

This tells us that, for example, the function $f(x, y, z) = x \cdot y \cdot z$ has partial derivatives

$$\begin{aligned} \partial_1 f(x, y, z) &= \frac{d}{dx} xyz = yz \\ \partial_2 f(x, y, z) &= \frac{d}{dy} xyz = xz \\ \partial_3 f(x, y, z) &= \frac{d}{dz} xyz = xy. \end{aligned}$$

Exercise 1.5. 1. Show that if f is differentiable, then all directional derivatives exist and fulfill $\frac{\partial}{\partial v} f(p) = Df(p)(v)$.

2. Conclude that $Df(p)$ is represented in basis $(x_0, \dots, x_n)$ by the **Jacobian matrix**

$$Jf(p) = \left(\frac{\partial}{\partial x_1} f(p), \dots, \frac{\partial}{\partial x_n} f(p) \right) = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(p) & \dots & \frac{\partial}{\partial x_n} f_1(p) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_1} f_n(p) & \dots & \frac{\partial}{\partial x_n} f_n(p) \end{pmatrix}$$

If the image of f is one-dimensional, the Jacobian matrix is simply a row vector. We call the transpose of this vector the **gradient of f at p** and denote it

$$\nabla f(p) := Jf(p)^\top = \begin{pmatrix} \frac{\partial}{\partial x_1} f(p) \\ \vdots \\ \frac{\partial}{\partial x_n} f(p) \end{pmatrix}$$

The transposition comes from the fact that we would like to be able to get the directional derivatives of f at p by taking a scalar product with the gradient, which gives us

$$\begin{aligned} \langle \nabla f(p), v \rangle &= \nabla f(p)^\top v \\ &= Jf(p) \cdot v \\ &= Df(p)(v) \\ &= \frac{\partial}{\partial v} f(p) \end{aligned}$$

There is one very important pitfall to watch out for when it comes to differentiability: We have just shown that if f is differentiable, we can get the differential by construction the Jacobian matrix from the partial derivatives. This makes it look like every partially differentiable function is differentiable, which turns out to not be true in general.

For example, there exist functions which are partially differentiable, but not continuous - we showed earlier that all differentiable functions are continuous, so these functions are quick counterexamples.

One example is

$$\begin{aligned} f : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{xy}{(x^2+y^2)^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \end{aligned}$$

This function has $\partial_1 f(0) = \partial_2 f(0) = 0$, since $f(h, 0) = f(0, h) = 0$ for all $h \in \mathbb{R}$, but isn't continuous, since $\lim_{n \rightarrow \infty} f\left(\frac{1}{n}, \frac{1}{n}\right) = \lim_{n \rightarrow \infty} \frac{n^2}{4} = \infty \neq f(0)$.

For a more involved counterexample, consider the function

$$\begin{aligned} f : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{y^3}{x^2+y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \end{aligned}$$

This function is continuous - for $(x, y) \neq 0$, it is a composition of continuous

functions, and thus continuous, and since $x^2 + y^2 \geq y^2$, we additionally have

$$\begin{aligned} |f(x, y)| &= \frac{|y^3|}{|x^2 + y^2|} \\ &= \frac{|y^3|}{x^2 + y^2} \\ &\leq \frac{|y^3|}{y^2} \\ &= |y|, \end{aligned}$$

so $f(x, y)$ converges to $f(0, 0) = 0$ as (x, y) converges to 0, implying that f is also continuous at the origin.

f is also directionally differentiable in all directions:

$$\begin{aligned} \partial_v f(0) &= \lim_{h \rightarrow 0} \frac{f(h \cdot v) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h \cdot v)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left(\frac{h^3 v_2^3}{h^2 v_1^2 + h^2 v_2^2} \right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^3 v_2^3}{h^3 v_1^2 + h^3 v_2^2} \\ &= \lim_{h \rightarrow 0} \frac{v_2^3}{v_1^2 + v_2^2} \\ &= \frac{v_2^3}{v_1^2 + v_2^2} \\ &= f(v) \end{aligned}$$

In particular, $\partial_1 f(0) = 0$ and $\partial_2 f(0) = 1$. However, f is not differentiable at 0, since otherwise our representation $Df(0) = Jf(0)$ would give us:

$$\begin{aligned} \lim_{\vec{h} \rightarrow 0} \frac{|f(h) - f(0) - Df(0)(\vec{h})|}{\|\vec{h}\|_2} &= \lim_{\vec{h} \rightarrow 0} \frac{|f(h) - Df(0)(h)|}{\|\vec{h}\|_2} \\ &= \lim_{\vec{h} \rightarrow 0} \left| \frac{f(h) - \partial_1 f(0) \cdot h_1 + \partial_2 f(0) \cdot h_2}{\sqrt{h_1^2 + h_2^2}} \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{f(h) - h_2}{\sqrt{h_1^2 + h_2^2}} \right| \\ &= \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3}{h_1^2 + h_2^2} - h_2 \right) \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2(h_1^2 + h_2^2)}{h_1^2 + h_2^2} \right) \right| \\ &= \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2 h_1^2 - h_2^3}{h_1^2 + h_2^2} \right) \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{-h_2 h_1^2}{h_1^2 + h_2^2} \right) \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3} \right| \end{aligned}$$

Since we are only interested in whether this expression converges to 0, we can

square it to simplify slightly:

$$\left| \frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3} \right|^2 = \left| \frac{h_2^2 h_1^4}{(h_1^2 + h_2^2)^3} \right|$$

If we now set $h := (\frac{1}{n}, \frac{1}{n})$, we get:

$$\left| \frac{h_2^2 h_1^4}{(h_1^2 + h_2^2)^3} \right| = \left| \frac{\frac{1}{n^6}}{\left(\frac{2}{n^2}\right)^3} \right| = \left| \frac{\frac{1}{n^6}}{\frac{8}{n^6}} \right| = \frac{1}{8} \neq 0$$

The important takeaway here is that directional differentiability only guarantees “differentiability along straight paths”, while full differentiability also requires “differentiability along curved paths”.

Theorem 1.6

$f : V \rightarrow W$ is continuously differentiable if and only if it is continuously partially differentiable.

Proof. $\Rightarrow$: We have $\text{Hom}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$. If the differential $Jf : \mathbb{R}^n \mapsto \mathbb{R}^{mn}$ exists, it is continuous if and only if every component $(Jf)_i : \mathbb{R}^n \mapsto \mathbb{R}$ is continuous. By the definition of Jf , these are the partial derivatives.

$\Leftarrow$: Since every normed complex vector space is isomorphic to a normed real vector space, we can assume without loss of generality that W is real. Since all norms are equivalent, and since a sequence in W converges iff. all components converge, we can additionally assume that W is one-dimensional, i.e. $W \cong \mathbb{R}$.

We need to show that

$$\lim_{h \rightarrow \vec{0}} \frac{f(p+h) - f(p) - Jf(p)h}{\|h\|_2} = \vec{0}$$

Let $h := \sum_{i=1}^n h_i e_i$. We decompose the path from p to $p+h$ into its partial steps, taking one coordinate direction at a time:

$$\forall k \in [0..n] :$$

$$p_k = p + \sum_{i=1}^k h_i e_i$$

Now, by the mean value theorem, there exists an $s_k \in [0, 1]$ such that

$$\begin{aligned} f(p_k) - f(p_{k-1}) &= f(p_{k-1} + h_k e_k) - f(p_{k-1}) \\ &= \partial_k f(p_{k-1} - s_k h_n e_n) h_k \end{aligned}$$

Now, since

$$Jf(p)(h) = \sum_{i=1}^n \partial_i f(p) h_i,$$

we get:

$$\begin{aligned}
 & \frac{|f(p+h) - f(p) - Jf(p)(h)|}{\|h\|_2} \\
 &= \frac{1}{\|h\|_2} \left| f(p_n) - f(p_0) - \sum_{i=1}^n \partial_i f(p) h_i \right| \\
 &= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n f(p_i) - f(p_{i-1}) - \partial_i f(p) h_i \right| \\
 &= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n \partial_i f(p_{i-1} + s_i h_i e_i) h_i - \partial_i f(p) h_i \right| \\
 &= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n (\partial_i f(p_{i-1} + s_i h_i e_i) - \partial_i f(p)) h_i \right| \\
 &= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n \left(\partial_i f \left(p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right) h_i \right| \\
 &\leq \frac{1}{\|h\|_2} \sum_{i=1}^n \left| \left(\partial_i f \left(p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right) h_i \right| \\
 &= \sum_{i=1}^n \left| \partial_i f \left(p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right| \cdot \frac{|h_i|}{\|h\|_2} \\
 &\leq \sum_{i=1}^n \left| \partial_i f \left(p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right|
 \end{aligned}$$

Since $s_i \in [0, 1]$, we have

$$\lim_{h \rightarrow 0} p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i = p$$

And thus, since the partial derivatives $\partial_i f$ are continuous, the right side converges to zero, so f is differentiable.

□

Exercise 1.7. Let V be an inner product space over a field $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$. Let $v \in V$. Then the function

$$\begin{aligned}
 \langle \cdot, v \rangle : V &\rightarrow \mathbb{F} \\
 w &\mapsto \langle w, v \rangle
 \end{aligned}$$

is differentiable, with $D \langle \cdot, v \rangle (w) = v^\top$ for all $w \in V$.

Exercise 1.8. Let b be a symmetric bilinear form. Let

$$q(v) := b(v, v)$$

be the corresponding quadratic form. Then q is differentiable, with

$$Dq(v)(w) = 2b(v, w)$$

1.2 Rules of Differentiation

Exercise 1.9. Show that the differential operator D is linear, i.e. that we have

$$D(\alpha f + \beta g)(p) = \alpha Df(p) + \beta Dg(p)$$

for f, g differentiable at p .

Theorem 1.10: Chain Rule in $\mathbb{R}^n$

Let $f : U \rightarrow \mathbb{R}^m$ and $g : V \rightarrow \mathbb{R}^r$ with $U \subset \mathbb{R}^n, V \subset \mathbb{R}^m$ open and $f(U) \subseteq V$. Let f be differentiable at a point $p \in U$ and let g be differentiable at $f(p)$. Then

$$D(g \circ f)(p) = Dg(f(p)) \circ Df(p)$$

Exercise 1.11. Verify that this version of the chain rule agrees with the familiar chain rule for $f, g : \mathbb{R} \rightarrow \mathbb{R}$.

Proof. By definition of the differential, we have:

$$\begin{aligned} f(x) &= f(p) \\ &\quad + Df(p)(x - p) \\ &\quad + \varepsilon_f(x)\|x - p\| \\ g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x) - f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x) - f(p)\| \end{aligned}$$

For some functions $\varepsilon_f : U \rightarrow \mathbb{R}^m, \varepsilon_g : U \rightarrow \mathbb{R}^r$ which are continuous in p and $f(p)$ and fulfill $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$.

The second equation tells us:

$$\begin{aligned} g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x) - f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x) - f(p)\| \end{aligned}$$

Plugging in our equation for $f(x)$ now lets us regroup our terms into an

equation of the same form as the definition of the differential:

$$\begin{aligned}
 g(f(x)) &= g(f(p)) \\
 &\quad + Dg(f(p)) \left[\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\| - f(p)}^{f(x)} \right] \\
 &\quad + \varepsilon_g(f(x)) \left\| \overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\| - f(p)}^{f(x)} \right\| \\
 &= g(f(p)) \\
 &\quad + Dg(f(p)) \left[Df(p)(x-p) + \varepsilon_f(x)\|x-p\| \right] \\
 &\quad + \varepsilon_g(f(x)) \left\| Df(p)(x-p) + \varepsilon_f(x)\|x-p\| \right\| \\
 &= g(f(p)) \\
 &\quad + Dg(f(p))Df(p)(x-p) \\
 &\quad + Dg(f(p))\varepsilon_f(x)\|x-p\| \\
 &\quad + \varepsilon_g(f(x)) \left\| Df(p)(x-p) + \varepsilon_f(x)\|x-p\| \right\| \\
 &= g(f(p)) \\
 &\quad + Dg(f(p))Df(p)(x-p) \\
 &\quad + \underbrace{\left(\varepsilon_f(x)Dg(f(p)) + \varepsilon_g(f(x)) \left\| Df(p) \left(\frac{x-p}{\|x-p\|} \right) + \varepsilon_f(x) \right\| \right)}_{:=\varepsilon_{g \circ f}(x)} \|x-p\| \\
 &= g(f(p)) \\
 &\quad + (Dg(f(p)) \circ Df(p))(x-p) \\
 &\quad + \varepsilon_{g \circ f}(x)\|x-p\|
 \end{aligned}$$

All that remains is to show that $\varepsilon_{g \circ f}(x)$ is continuous if we set

$$\varepsilon_{g \circ f}(p) = 0.$$

Since we have $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$, we have our desired result if we can show that

$$\left\| Df(p) \left(\frac{x-p}{\|x-p\|} \right) \right\| = \frac{\|Df(p)(x-p)\|}{\|x-p\|}.$$

is bounded. Since $\mathbb{R}^n$ is finite-dimensional, we can find a constant c such that

$$\|Df(p)(x-p)\| \leq c \cdot \|x-p\|,$$

which means that our quotient is indeed bounded, completing the proof. $\square$

Theorem 1.12: Product Rule in $\mathbb{R}^n$

Let $U \subseteq \mathbb{R}^n$. Let $f, g : U \rightarrow \mathbb{R}$ at a point $p \in U$. Then

$$D(f \cdot g)(p) = Df(p) \cdot g(p) + f(p) \cdot Dg(p).$$

Exercise 1.13. Verify that this version of the product rule agrees with the familiar product rule for $f, g : \mathbb{R} \rightarrow \mathbb{R}$.

Proof. We define the multiplication function $m(x, y) := x \cdot y$. The partial derivatives are $\partial_1 m(x, y) = y$ and $\partial_2 m(x, y) = x$, i.e.

$$Dm(x, y) = \begin{pmatrix} y & x \end{pmatrix}$$

Applying the chain rule now gives

$$\begin{aligned} D(f \cdot g)(p) &= D(m \circ (f, g))(p) \\ &= Dm((f, g)(p)) \circ D(f, g)(p) \\ &= Dm(f(p), g(p)) \circ (Df(p), Dg(p)) \\ &= Dm(f(p), g(p)) \cdot (Df(p), Dg(p)) \\ &= \begin{pmatrix} g(p) & f(p) \end{pmatrix} \cdot \begin{pmatrix} Df(p) \\ Dg(p) \end{pmatrix} \\ &= Df(p) \cdot g(p) + f(p) \cdot Dg(p). \end{aligned}$$

□

Exercise 1.14. Let $f : V \rightarrow W$ be differentiable. Let $\gamma : [0, 1] \rightarrow V$ be differentiable. Then $f \circ \gamma$ is differentiable with

$$(f \circ \gamma)'(t) = Df(\gamma'(t)) \cdot \gamma'(t)$$

Theorem 1.15. Let $U \subseteq V$ be open. Let $f : U \rightarrow W$ be differentiable with $Df = 0$. Then f is locally constant.

Corollary 1.16. Let $U \subseteq V$ be open and connected. Let $f : U \rightarrow W$ be differentiable with $Df = 0$. Then f is constant.

1.3 Higher Partial Derivatives

Theorem 1.17: Schwartz's Theorem

Let $f : U \subseteq V \rightarrow W$ be twice continuously partially differentiable. Then the order of partial derivation does not matter, i.e.

$$\partial_i \partial_j f = \partial_j \partial_i f$$

for all $i, j \in [1..n]$.

Theorem 1.18: First-Order Taylor Expansion in multiple Variables

Let $f : U \subseteq V \rightarrow W$. Let f be continuously differentiable. Let $x \in U$, $h \in V$ fulfill $[x, x+h] \subseteq U$. Then

$$f(x+h) = f(x) + Df(x)(h) + r(h),$$

with $r \in o(\|h\|)$.

1.4 Differential Operators

Chapter 2

The Two Queens of Multivariable Calculus

2.1 The Inverse Function Theorem

Theorem 2.1: Inverse Function Theorem

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \rightarrow \mathbb{R}^n$ be continuously differentiable. Let $x \in U$ be a point such that $Df(x) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible. Then there exists an open neighbourhood V of x such that:

1. $f(V)$ is an open neighbourhood of $f(x)$.
2. $f|_V$ is a C^1 -diffeomorphism, i.e. invertible with continuously differentiable inverse.

In short: if a continuously differentiable function has an invertible differential at a given point, there exists an open neighbourhood around that point in which the function is invertible:

Corollary 2.2. Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \rightarrow \mathbb{R}^n$ be continuously differentiable. Let $Df(x)$ be invertible for all $x \in \mathbb{R}^n$. Then $f(U)$ is open.

2.2 The Implicit Function Theorem

Theorem 2.3: Implicit Function Theorem

Let $U \subseteq \mathbb{R}^m \times \mathbb{R}^k$ be open. Let $f : U \rightarrow \mathbb{R}^k$ be continuously differentiable. Let $(x, y) \in U$ with $f(x, y) = z$. Let $Df|_{\mathbb{R}^k}(x, y) : \mathbb{R}^k \rightarrow \mathbb{R}^k$ be invertible. Then there exist

1. open neighbourhoods V of x and W of y ,
2. and a continuously differentiable function $g : V \rightarrow W$,

such that the set of values in $V \times W$ for which f takes the value z is the graph of g , i.e.

$$\{(v, w) \in V \times W \mid f(v, w) = z\} = \{(v, g(v)) \mid v \in V\}$$

In short: under mild assumptions, the level sets of a continuously differentiable function are graphs of continuously differentiable functions:

Chapter 3

Multiple Integrals

Lemma 3.1. Let $K \subseteq \mathbb{R}^n$ be compact. Let $f : K \times [a, b] \rightarrow \mathbb{R}$ be continuous. Then the function

$$x \mapsto \int_a^b f(x, t) dt$$

is uniformly continuous.

Proof. $K \times [a, b]$ is a finite product of compact sets, i.e. compact by Theorem 2.32. Thus, f is uniformly continuous by Theorem 2.5, which means that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any $x, y \in \mathbb{R}, t \in \mathbb{R}^n$

$$d((x, t), (y, t)) < \delta \implies d(f(x, t), f(y, t)) < \varepsilon.$$

This tells us that for any $x, y \in \mathbb{R}$,

$$\begin{aligned} d(x, y) < \delta &\implies d((x, t), (y, t)) < \varepsilon \\ &\implies |f(x, t) - f(y, t)| = d(f(x, t), f(y, t)) < \varepsilon \\ &\implies d\left(\int_a^b f(x, t) dt, \int_a^b f(y, t) dt\right) \\ &= \left|\int_a^b f(x, t) dt - \int_a^b f(y, t) dt\right| \\ &\leq \int_a^b |f(x, t) - f(y, t)| dt \\ &\leq (b - a)\varepsilon, \end{aligned}$$

since ε was arbitrary, our function is uniformly continuous as desired. $\square$

Definition 3.2. Given a compact cuboid $Q = \prod_{i=1}^n [a_i, b_i] \subseteq \mathbb{R}^n$ and a continuous function $f : Q \rightarrow \mathbb{R}$, we declare the *integral of f over Q* to be the nested integral

$$\int_Q f(x) d^n x := \int_{a_n}^{b_n} \dots \int_{a_1}^{b_1} f(x_1, \dots, x_n) dx_1 \dots dx_n$$

By Lemma 3.1, all involved functions are continuous, and thus their integrals are indeed well-defined. This definition will become a fully justified theorem

once we begin our study of the Lebesgue integral later on.

Exercise 3.3. Verify that linearity and monotonicity carry over from the single-variable case, i.e show that for any $f, g : Q \rightarrow \mathbb{R}$ and $\lambda \in \mathbb{R}$, we have:

1. $\int_Q f(x) + g(x) d^n x = \int_Q f(x) d^n x + \int_Q g(x) d^n x$
2. $\int_Q \lambda f(x) d^n x = \lambda \int_Q f(x) d^n x$
3. $f \leq g \implies \int_Q f(x) d^n x \leq \int_Q g(x) d^n x$
4. $\left| \int_Q f(x) d^n x \right| \leq \int_Q |f(x)| d^n x$

Definition 3.4. We define

$$\text{vol } Q = \int_Q 1 = \prod_{i=1}^n |b_i - a_i|$$

Definition 3.5. By partitioning the intervals $[a, b] \times [c, d] := Q \subseteq \mathbb{R}^2$ into r equidistant chunks with boundaries

$$\begin{aligned} a &= a_1 \leq a_2 \leq \dots \leq a_r = b \\ c &= c_1 \leq c_2 \leq \dots \leq c_r = d, \end{aligned}$$

we partition Q into r^2 equally sized smaller rectangles

$$Q_{ij} := [a_i, c_{i+1}] \times [a_j, c_{j+1}]$$

which must each have area $\text{vol } Q / r^2$. We can thus define *two-dimensional Riemann sums* analogously to how we did it in one dimension:

$$S_r^2(f) := \sum_{i=0}^{r-1} \sum_{j=0}^{r-1} f(a_i, c_j) \frac{\text{vol } Q}{r^2}$$

Lemma 3.6: Multivariable Integration via Riemann Sums

Given a compact cuboid $Q \subseteq \mathbb{R}^2$ and a continuous function $f : Q \rightarrow \mathbb{R}$, we have

$$\int_Q f(x) d^2 x = \lim_{r \rightarrow \infty} S_r^2(f)$$

Proof. Applying the single-variable version of this theorem twice, we get:

$$\begin{aligned} \int_Q f(x) d^2 x &= \int_c^d \int_a^b f(x_1, x_2) dx_1 dx_2 \\ &= \lim_{r \rightarrow \infty} S_r \left(\int_a^b f(x_1, x_2) dx_1 \right) \\ &= \lim_{r \rightarrow \infty} S_r^2(f(x_1, x_2)) \end{aligned}$$

□

Chapter 4

Curves

Chapter 5

Surfaces

Part IV

Point-Set Topology

Chapter 1

Topological Spaces

1.1 Topologies

Definition 1.1: Accumulation points and isolated points

Let X be a topological space. Let $A \subseteq X$. Then we call $x \in X$ an *accumulation point* of A if every neighbourhood of x contains a point in A other than x itself.

If a point $x \in A$ isn't an accumulation point of A , we call x an *isolated point*.

Definition 1.2. Let X be a topological space. Let $A \subseteq X$. Then the set of all accumulation points of A is denoted A' , and is known as the *derived set of A* .

Exercise 1.3. Show that $A' \subseteq \overline{A}$.

1.2 Basic Separation Properties

A very common axiom is the Hausdorff separation axiom, which says that any two distinct points can be housed orf from each other by disjoint open sets.

“Topology via Logic”,
Steven Vickers

Definition 1.4. Let $(X, \mathcal{O})$ be a topological space. Let $x, y \in X$. Then we call x and y *topologically distinguishable* iff. $\mathcal{U}(x) \neq \mathcal{U}(y)$, i.e. one of the points has a neighborhood not containing the other.

Definition 1.5: T_0 Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ *T_0* if every pair of points is topologically distinguishable.

Definition 1.6: T_1 Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ *T_1* if for every pair of points, both of them have neighbourhoods not containing the other:

$$\begin{aligned} \forall x, y \in X : \\ \exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) : \\ x \notin U_y, y \notin U_x. \end{aligned}$$

Exercise 1.7. 1. Find a topological space that isn't T_0 .

2. Show that every T_1 space is T_0 .

3. Find a space that is T_0 , but not T_1 .

Exercise 1.8. Show that $(X, \mathcal{O})$ is T_1 if and only if every singleton set is closed.

Exercise 1.9. Show that $(X, \mathcal{O})$ is T_1 if and only if every subset of X is the intersection of all of its neighborhoods:

$$\forall Y \subseteq X : Y = \bigcap \{U \mid U \in \mathcal{U}(Y)\}$$

Definition 1.10: Hausdorff Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ *T_2* or *Hausdorff* if every pair of points can be housed orf from each other, i.e. they have disjoint neighbourhoods:

$$\begin{aligned} \forall x \neq y \in X : \\ \exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) : \\ U_x \cap U_y = \emptyset \end{aligned}$$

Exercise 1.11. Show that every Hausdorff space is T_1 .

Exercise 1.12. Let $(X, \mathcal{O})$ be Hausdorff. Then every convergent sequence in X has a unique limit.

Exercise 1.13. Let X be a Hausdorff space. Let x be an accumulation point of $A \subseteq X$. Then every neighbourhood of x contains infinitely many points of A .

Lemma 1.14. Let X be a Hausdorff space. Then the set of accumulation points of any subset is closed.

Proof. Let $A \subseteq X$. Let A' be the set of accumulation points of A . Let x be an accumulation point of A' . Since X is Hausdorff, every neighbourhood of x must contain infinitely many points $x_i \in A'$. But since these neighbourhoods are also neighbourhoods of the x_i , they must also all contain infinitely many points in A .

Therefore, x is already an accumulation point of A , i.e. $x \in A'$. Therefore, A' contains all of its accumulation points, and is thus closed. $\square$

Exercise 1.15. Find a space X and a subset A such that the set of accumulation points of A isn't closed.

Theorem 1.16. Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if every point is the intersection of the closures of all of its neighbourhoods:

$$\forall x \in X : \{x\} = \bigcap \left\{ \overline{U} \mid U \in \mathcal{U}(x) \right\}$$

Proof. $\Rightarrow$ Let $(X, \mathcal{O})$ be Hausdorff. Let $x, y \in X$ and $y \notin \{x\}$, i.e. $y \neq x$. Then, by Hausdorff, there exist disjoint neighborhoods U_x and U_y of x and y . Since U_x and U_y are disjoint, we have $U_x \subseteq X \setminus U_y$. But then, by monotonicity of the closure operator, we get

$$\overline{U_x} \subseteq \overline{X \setminus U_y} = X \setminus U_y,$$

i.e. $y \notin \overline{U_x}$, i.e.

$$y \notin \bigcap \left\{ \overline{U} \mid U \in \mathcal{U}(x) \right\}.$$

$\Leftarrow$ Let $x \neq y$. Let $\{x\} = \bigcap \left\{ \overline{U} \mid U \in \mathcal{U}(x) \right\}$. Then there must be an open neighborhood U_x of x whose closure doesn't contain y . But then the complement $U_y := X \setminus \overline{U_x}$ of that closure is an open neighborhood of y that is disjoint from U_x .

$\square$

1.3 Bases and Countability

Definition 1.17

We call a topological space *first countable* if it has a countable neighbourhood basis around every point x , i.e. a set of neighbourhoods B_i of x such that every neighbourhood of x contains some B_i .

Exercise 1.18. Every metric space is first countable.

It turns out that first countable spaces are exactly the spaces where sequences are sufficient to maintain our intuitions about how sequence convergence interacts with topological properties:

Theorem 1.19: Detecting openness and closedness using sequences

Let X be first countable. Let $A \subseteq X$. Let $x \in X$. Then:

1. $x \in \text{int } A$ if and only if every sequence in X converging to x is eventually in A .
2. $x \in \overline{A}$ if and only if there exists a sequence in A converging to x .

Proof. Copy the proof of the metric case, replacing open balls with our countable neighbourhood basis. $\square$

Corollary 1.20. Let X be first countable. Let $A \subseteq X$. Let $x \in X$. Then:

1. A is closed in X if and only if it contains every limit of every convergent sequence of points in A .
2. A is open in X if and only if every sequence in X converging to a point of A is eventually in A .

Definition 1.21

We call a topological space *separable* if it contains a countable dense subset.

Exercise 1.22.

1. Every countable space is separable.
2. $\mathbb{R}^n$ is separable for every $n \in \mathbb{N}$.

Definition 1.23

We call a topological space *second countable* if it has a countable basis.

Exercise 1.24. $\mathbb{R}^n$ is second countable for every $n \in \mathbb{N}$.

Exercise 1.25. Let X be second countable. Then:

1. X is first countable.
2. X is separable.

Theorem 1.26. A metric space is second countable if and only if it is separable.

Chapter 2

New Spaces from Old

2.1 The Subspace Topology

Theorem 2.1

Let $(X, \mathcal{O})$ be a topological space, and let $S \subseteq X$ be any subset. Then the collection

$$\mathcal{O}_S := \{U \subseteq S \mid \exists V \in \mathcal{O} : U = S \cap V\},$$

consisting of all intersections of open sets of X with S , forms a topology, which we call the **subspace topology**.

Exercise 2.2. Let $S \subseteq X$. Show that $B \subseteq S$ is closed in S if and only if it is equal to the intersection of S with some closed subset of X .

Exercise 2.3. Let X be a topological space. Let $U \subseteq S \subseteq X$.

1. If U is open in X , then U is open in S .
2. If U is closed in X , then U is closed in S .

Exercise 2.4. Let X be a topological space. Let $U \subseteq S \subseteq X$.

1. If U is open in S , and S is open in X , then U is open in X .
2. If U is closed in S , and S is closed in X , then U is closed in X .

If $U \subseteq S \subseteq X$, the notations $\overline{U}$, $\text{int } U$, and $\text{ext } U$ will always refer to the closure, interior, and exterior of U in X , not in S .

Exercise 2.5. Let X be a topological space. Let $U \subseteq S \subseteq X$.

1. The closure of U in S is equal to $\overline{U} \cap S$.
2. Find an example where the interior of U in S is not equal to $\text{int } U \cap S$.
3. Show that $\text{int } U \cap S$ is a subset of the interior of U in S .

Theorem 2.6: Universal Property of the Subspace Topology

Let X and Y be topological spaces. Let $S \subseteq Y$. Then $f : X \rightarrow S$ is continuous if and only if the embedded map $\iota_S \circ f : X \rightarrow Y$ is continuous.

Corollary 2.7. Let $S \subseteq X$. Then the inclusion map $\iota_S : S \hookrightarrow X$ is continuous.

Corollary 2.8. Let X and Y be topological spaces. Let $f : X \rightarrow Y$ be continuous. Then:

1. If $S \subseteq X$, the restriction

$$f|_S : S \rightarrow Y$$

is continuous.

2. If $f(X) \subseteq T \subseteq Y$, the map

$$f : X \rightarrow T$$

is well-defined and continuous.

3. If $Y \subseteq Z$, then the embedded function

$$\iota_Y \circ f : X \rightarrow Z$$

is continuous.

Exercise 2.9. Let X be a topological space. Let $R \subseteq S \subseteq X$. Then the subspace topologies of R in S and X agree.

Exercise 2.10. Let $(X, \mathcal{O})$ be a topological space. Let $S \subseteq X$. Then if $\mathcal{B}$ is a basis of $\mathcal{O}$, the collection

$$\mathcal{B}_S = \{B \cap S \mid B \in \mathcal{B}\}$$

is a basis of the subspace topology $\mathcal{O}_S$.

Exercise 2.11.

1. Every subspace of a T_0 space is T_0 .
2. Every subspace of a T_1 space is T_1 .
3. Every subspace of a Hausdorff space is Hausdorff.
4. Every subspace of a first countable space is first countable.
5. Every subspace of a second countable space is second countable.

Definition 2.12

A map that is a homeomorphism onto its image is called a **topological embedding**, or just **embedding**.

Exercise 2.13. A homeomorphism is a surjective topological embedding.

Exercise 2.14. Let X be a topological space. Let $S \subseteq X$. Then the inclusion map $\iota_S : S \hookrightarrow X$ is an embedding.

Exercise 2.15. Every embedding is continuous and injective.

Exercise 2.16. Every injective continuous map that is open or closed is an embedding.

Exercise 2.17. Find an embedding that is neither open nor closed.

Theorem 2.18

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \rightarrow \mathbb{R}^k$ be continuous. Then the graph $\Gamma(f)$ of f is homeomorphic to U .

Proof. Let $\Phi_f : U \rightarrow \mathbb{R}^{n+k}$ be the canonical map

$$\Phi_f(x) = (x, f(x))$$

taking U to its graph.

By definition of the graph $\Gamma(f)$, $\Phi_f(x) : U \rightarrow \Gamma(f)$ is surjective.

Since Φ_f is a composition of continuous maps, it is continuous.

Since the first n components of Φ_f are the identity, Φ_f is injective.

Finally, the projection $p : \mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$ is continuous, so the restriction $p|_{\Gamma(f)}$ is a continuous inverse of Φ_f . $\square$

2.2 Product Spaces

Lemma 2.19. Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if the diagonal

$$\Delta := \{(x, x) \mid x \in X\}$$

is closed in the product topology.

Proof. $\Rightarrow$ Assume $(X, \mathcal{O})$ is Hausdorff. Let $(x, y) \notin \Delta$. Then $x \neq y$ by definition of Δ . Therefore, there exist disjoint neighbourhoods U_x, U_y of x and y . Since these neighborhoods are disjoint, we have

$$\Delta \cap (U_x \times U_y) = \emptyset$$

. Therefore, every point $(x, y) \notin \Delta$ has an open neighborhood $(U_x \times U_y)$ that doesn't intersect Δ , i.e. Δ is closed.

$\Leftarrow$ Let $x \neq y$. Assume that Δ is closed. Then $(x, y) \notin \Delta$ has a basic open neighborhood $U_x \times U_y$ that doesn't intersect Δ . Since $U_x \times U_y$ doesn't intersect Δ , U_x and U_y must be disjoint.

□

2.3 Disjoint Union Spaces

2.4 Quotient Spaces

2.5 Adjunction Spaces

Chapter 3

Topological Manifolds

Definition 3.1

We call a topological space X *locally Euclidean of dimension n* if every point $x \in X$ has a neighbourhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Exercise 3.2. If a topological space X is a locally euclidean of dimension n , then:

1. Every point $x \in X$ has a neighbourhood homeomorphic to an open ball in $\mathbb{R}^n$, or
2. Every point $x \in X$ has a neighbourhood homeomorphic to $\mathbb{R}^n$, or

Exercise 3.3. Show that a topological space is locally Euclidean of dimension 0 if and only if it is discrete.

Definition 3.4

A topological space is called an *n -dimensional topological manifold*, or just *topological n -manifold*, if it is:

1. locally euclidean of dimension n ,
2. second countable,
3. and Hausdorff.

The latter two requirements exclude some pathological spaces - the long line would be a topological manifold if it was second countable, and the bug-eyed line would be a topological manifold if it was Hausdorff. Most interesting results about topological manifolds require both of these properties, so little would be gained by omitting them.

Intuitively, it sounds very sensible that a nonempty topological space cannot be both a topological m -manifold and a topological n -manifold. This statement turns out to be true - however, we will save this proof for later, since it requires a substantial amount of machinery borrowed from algebraic topology.

Lemma 3.5. Every open subset of a topological n -manifold is a topological n -manifold.

Exercise 3.6. A topological space is a topological 0-manifold if and only if it is countable and discrete.

Theorem 3.7. Any separable metric space that is locally Euclidean of dimension n is a topological n -manifold.

Definition 3.8

We call a topological space an *n -dimensional manifold with boundary* if it is second countable, Hausdorff, and every open point has a neighbourhood homeomorphic to an open subset of either $\mathbb{R}^n$, or to the n -dimensional upper half-space

$$\mathbb{H}^n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$$

Definition 3.9

Let M be a topological n -manifold with boundary. Then a *coordinate chart for M* is a pair (U, φ) , consisting of an open subset $U \subseteq M$, called a *coordinate domain*, and a homeomorphism φ from U to an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, called a *coordinate map*. If U is homeomorphic to an open ball, we call U a *coordinate ball*.

Note that if we can assign coordinate maps for some covering of a topological space X , then that space must by definition be locally euclidean.

Definition 3.10: Manifold interior and manifold boundary

If $\varphi(U)$ is an open subset of $\mathbb{R}^n$, we call φ an *interior chart* and points $p \in U$ *interior points*. If $\varphi(U)$ is a subset of $\mathbb{H}^n$ with $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$, we call φ a *boundary chart*. If there exists a boundary chart taking p to $\partial\mathbb{H}^n$, we call p a *boundary point of M* .

Every manifold is a manifold with boundary, but the opposite might not hold. We sometimes use the redundant term "manifold without boundary" to refer to a manifold with boundary which has the empty set as its boundary. The word "manifold", without further qualifiers, always means a manifold without boundary.

The definition of a manifold's boundary only agrees with the topological definition once we find the right ambient space to embed M into, since M always has empty topological boundary when viewed as a topological space of its own.

For example, the manifold boundary of the closed unit disk $\overline{\mathbb{B}}^2$ is the unit circle, which agrees with the topological boundary if the disk is viewed as a subset of $\mathbb{R}^2$, but not with the topological boundary of the disk when viewed as its own topological space (which is trivially empty, as stated), or with the topological boundary of the disk viewed as a subset of $\mathbb{R}^3$ (which is the entire disk).

Theorem 3.11. If M is a topological n -manifold with boundary, then $\text{int } M$ is both an open subset of M and a topological n -manifold without boundary.

Proof. Since $\text{int } M$ is a union of coordinate domains, it is open. Second-countability and Hausdorffness are trivially inherited by subspaces. By definition, $\text{int } M$ consists only of points homeomorphic to an open subset of $\mathbb{R}^n$, so $\text{int } M$ is additionally locally euclidean of dimension n . $\square$

Proposition 3.12. Let M be a topological manifold with boundary. Then a point $x \in M$ cannot be both a boundary point and an interior point.

The general proof once again requires some tricky algebraic topology, so we save it for later.

Corollary 3.13. If M is a nonempty topological n -manifold with boundary, then ∂M is a closed subset of M , and M is a topological n -manifold if and only if $\partial M = \emptyset$.

Proof. Since $\text{int } M$ is an open subset of M , $\partial M = M \setminus \text{int } M$ is closed. If M is a manifold, then every point is an interior point, so $M = \text{int } M$, so $\partial M = \emptyset$, and vice versa. $\square$

3.1 Manifolds from Embeddings

Theorem 3.14

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \rightarrow \mathbb{R}^k$ be continuous. Then the graph $\Gamma(f)$ of f is a topological n -manifold.

Proof. By 2.18, $\Gamma(f)$ is homeomorphic to $U \subseteq \mathbb{R}^n$, making $\Gamma(f)$ a trivial topological n -manifold coverable by a single coordinate domain. $\square$

Theorem 3.15

The unit n -spheres $\mathbb{S}^n$ are topological n -manifolds.

Proof. For each $i \in [1 \dots n + 1]$, let U_i^+ and U_i^- denote the sets

$$U_i^+ := \{x \in \mathbb{R}^{n+1} \mid x_i > 0\}$$

$$U_i^- := \{x \in \mathbb{R}^{n+1} \mid x_i < 0\},$$

which are open in $\mathbb{R}^{n+1}$.

Since $\mathbb{S}^n$ doesn't contain the origin, every point on $\mathbb{S}^n$ must have at least one non-zero coordinate, so these sets cover $\mathbb{S}^{n+1}$.

On these sets, $\|x\| = 1$ has a unique solution, so we can find the intersection $x \in \mathbb{S}^n \cap U_i^\pm$ by solving for x_i , which gives us

$$\begin{aligned} \sqrt{\sum_{j=1}^{n+1} x_j^2} &= 1 \\ \Leftrightarrow \sum_{j=1}^{n+1} x_j^2 &= 1 \\ \Leftrightarrow x_i^2 &= 1 - \sum_{j \neq i} x_j^2 \\ \Leftrightarrow x_i &= \begin{cases} \sqrt{1 - \sum_{j \neq i} x_j^2} & x \in U_i^+ \\ -\sqrt{1 - \sum_{j \neq i} x_j^2} & x \in U_i^- \end{cases} \end{aligned}$$

We can thus describe the intersections of $\mathbb{S}^n$ with our sets $U_i^\pm$ by taking every component except the i -th to be arbitrary, and setting the i -th component to the value taken by these continuous square root functions. Swapping the i -th and the $n + 1$ -st components is continuous, so our intersections are the graphs of these functions. Since these intersections cover $\mathbb{S}^n$, and our functions are $\mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbb{S}^n$ is a topological n -manifold. $\square$

Theorem 3.16. Let $p \in \mathbb{S}^n$. Let $H := \langle p \rangle^\perp$ be the hyperplane given as the orthog-

onal complement of the line through p and the origin. Then the *stereographic projection map* σ_p taking a point $x \in \mathbb{S}^n \setminus \{p\}$ to the point where the line $\vec{p}x$ intersects H , is a homeomorphism $\mathbb{S}^n \setminus \{p\} \cong H(\cong \mathbb{R}^n)$

Proof. Let $N := e_{n+1} = (0, \dots, n+1)^\top$ be the "north pole" of the sphere. Take a rotation R in the plane spanned by p and N mapping p to N . Since rotations in the origin preserve $\mathbb{S}^n$, we get

$$\sigma_p = R^{-1} \circ \sigma_N \circ R.$$

Since rotations are orthogonal maps, and thus homeomorphisms, it suffices to show that σ_N is a homeomorphism.

In this case, the hyperplane H orthogonal to $\langle N \rangle$ is simply given by $x_{n+1} = 0$. Therefore, the intersection $u := \sigma_N(x)$ of the line $\vec{N}x = \{\lambda.(x - N) + N \mid \lambda \in \mathbb{R}\}$ can be found by finding the value of λ that makes the $n+1$ st component zero. We get:

$$\begin{aligned} \lambda.(x_{n+1} - 1) + 1 &= 0 \\ \Leftrightarrow \lambda.(x_{n+1} - 1) &= -1 \\ \Leftrightarrow \lambda &= \frac{1}{1 - x_{n+1}} \end{aligned}$$

For $i \neq n+1$, we have $N_i = 0$, so $u_i = \lambda.x_i$. Substituting our λ into this equation gives

$$\sigma_N(x) = \frac{x_1, \dots, x_n, 0}{1 - x_{n+1}}$$

This map is continuous for $x_{n+1} \neq 1$, i.e. $x \in \mathbb{S}^n \setminus \{N\}$.

Solving our equations for x instead gives

$$\begin{aligned} x_i &= \frac{u_i}{\lambda}, \\ x_{n+1} &= \frac{\lambda - 1}{\lambda}, \end{aligned}$$

And since our preimage needs to lie on $\mathbb{S}^n$, we substitute into $\|x\|^2 = 1$ and get

$$\begin{aligned} 1 &= \sum_{i=1}^{n+1} x_i^2 \\ &= \left(\sum_{i=1}^n \left(\frac{u_i}{\lambda} \right)^2 \right) + \left(\frac{\lambda - 1}{\lambda} \right)^2 \\ &= \frac{\sum_{i=1}^n u_i^2}{\lambda^2} + \frac{\lambda^2 - 2\lambda + 1}{\lambda^2} \\ &= \frac{\|u\|^2 + \lambda^2 - 2\lambda + 1}{\lambda^2} \\ \Leftrightarrow \lambda^2 &= \|u\|^2 + \lambda^2 - 2\lambda + 1 \\ \Leftrightarrow 2\lambda &= \|u\|^2 + 1 \\ \Leftrightarrow \lambda &= \frac{\|u\|^2 + 1}{2} \end{aligned}$$

Substituting back into our two equations gives

$$\sigma_N^{-1}(u_1, \dots, u_n) = \frac{(2u_1, \dots, 2u_n, \|u\|^2 - 1)^\top}{\|u\|^2 + 1},$$

which is continuous. Therefore, $\sigma_N : \mathbb{S}^n \setminus \{N\} \cong H \cong \mathbb{R}^n$ is indeed a homeomorphism, implying that every σ_p is a homeomorphism. $\square$

σ_N and σ_S (where $S = -N$ is the "south pole") give an alternate coordinate atlas on $\mathbb{S}^n$, which is sometimes easier to work with than our original atlas given by square root functions.

Theorem 3.17. The unit torus, given by the equations

Chapter 4

Topologies on Algebraic Structures

4.1 Topological Groups

Definition 4.1

Let $(G, \circ)$ be a group. Let $\mathcal{O}$ be a topology on G . Then we call $(G, \circ, \mathcal{O})$ a *topological group* iff. the maps $^{-1} : g \rightarrow g^{-1}$ and $\circ : G \times G \rightarrow G$ are continuous.

Exercise 4.2. Show that $(G, \circ, \mathcal{O})$ is a topological group if and only if the map

$$\varphi : (g, h) \mapsto g \circ h^{-1}$$

is continuous.

Exercise 4.3. Let $(G, \circ, \mathcal{O})$ be a topological group. Then for any $g \in G$, the left and right group operation maps $l_g : h \mapsto g \circ h$ and $r_g : h \mapsto h \circ g$ are homeomorphisms.

Exercise 4.4. Let $(G, \circ, \mathcal{O}_G)$ be a topological group. Let $(H, \circ, \mathcal{O}_H)$ be a normal subgroup equipped with the subspace topology, and let $(G/H, \circ, \mathcal{O}_q)$ be the quotient group, equipped with the quotient topology. Then the quotient map $p : G \rightarrow G/H$ is continuous and open.

Corollary 4.5. If H is closed in G and G is Hausdorff, then G/H is Hausdorff. If G/H is T_1 , then H is closed.

Chapter 5

Connectedness

Chapter 6

Compactness

The familiar notions of open coverings and compactness generalize directly from metric spaces to topological spaces. The most major difference is that sequential compactness is generally weaker than "covering compactness", and has to be replaced with a notion of compactness based on ultrafilter convergence.

Definition 6.1. Let $(X, \mathcal{O})$ be a topological space. Then we call a set $\mathcal{U} \subseteq \mathcal{O}$ of open subsets an *open covering* of X if every point of X is contained in the union of all elements of $\mathcal{U}$, i.e. if

$$\bigcup_{U_i \in \mathcal{U}} U_i = X$$

Definition 6.2: Compactness

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ *compact* if every open covering of X contains a finite open covering as a subset. We call a subset $K \subseteq X$ compact if it is compact as a topological space equipped with the subspace topology.

Exercise 6.3. Every closed subset of a compact space is compact.

Exercise 6.4. Let $(X, \mathcal{O})$ be compact and $f : X \rightarrow Y$ be continuous. Then $f[X]$ is compact.

Exercise 6.5. Let $(X, \mathcal{O})$ be a nonempty topological space. Then the following are equivalent:

1. $(X, \mathcal{O})$ is compact, i.e. every open covering has a finite subcovering.
2. Every set $\mathcal{A}$ of closed sets whose intersection is the empty set has a finite subset whose intersection is the empty set.
3. If $\mathcal{A}$ is a set of closed sets such that the intersection over any finite subset of $\mathcal{A}$ is non-empty, the intersection of $\mathcal{A}$ is non-empty.
4. Every filter on $(X, \mathcal{O})$ has a limit point.
5. Every ultrafilter on $(X, \mathcal{O})$ is convergent.

Theorem 6.6. (Alexander Subbase Theorem) Let $(X, \mathcal{O})$ be a topological space. Let $\mathcal{S}$ be a subbasis. Then X is compact if every open covering of X by sets of $\mathcal{S}$ has a finite subcovering.

Proof. Assume that X is not compact. Then there exists a non-convergent ultrafilter $\mathcal{F}$, i.e. we have

$$\begin{aligned} \forall x \in X : \\ \exists U_x \in \mathcal{U}(x) : \\ (X \setminus U_x) \in \mathcal{F} \end{aligned}$$

By definition of subbases, every $U_x \in \mathcal{U}(x)$ contains an open subset O_x of the form

$$O_x = \bigcap_{k=0}^n S_{k,x}$$

for some $S_{k,x} \in \mathcal{S}$ and $n \in \mathbb{N}$. We therefore have

$$\begin{aligned} X \setminus U_x &\subseteq X \setminus O_x \\ &= X \setminus \bigcap_{k=0}^n S_{k,x} \end{aligned}$$

which implies $X \setminus \bigcap_{k=0}^n S_{k,x} \in \mathcal{F}$ by the filter axioms. Since the intersection of this set with $\bigcap_{k=0}^n S_{k,x} \in \mathcal{F}$ is empty, and filters are closed under finite intersection, $\mathcal{F}$ cannot contain every $S_{k,x}$. Therefore, since $\mathcal{F}$ is an ultrafilter, it must contain at least one set $X \setminus S_{k,x}$ for every x . We will refer to these $S_{k,x}$ with $X \setminus S_{k,x} \in \mathcal{F}$ as S_x .

Since every S_x is an open neighborhood of x , and thus contains x , the set of all S_x is an open covering of X . By compactness, there must be a finite subcovering

$$\bigcup_{x \in X'} S_x = X.$$

Now, since $\mathcal{F}$ contains every $X \setminus S_x$, it also contains

$$\bigcap_{x \in X'} (X \setminus S_x) = X \setminus \bigcup_{x \in X'} S_x = X \setminus X = \emptyset.$$

which contradicts $\mathcal{F}$ being a filter. $\square$

Lemma 6.7. Let $(X, \mathcal{O})$ be a Hausdorff space. Let $K \subseteq X$ be compact. Then K can be separated from any point $x \in X \setminus K$ by open neighborhoods, i.e:

$$\begin{aligned} \forall x \in X : \\ \exists U_x \in \mathcal{U}(x), U_K \in \mathcal{U}(K) : \\ U_x \cap U_K = \emptyset \end{aligned}$$

Exercise 6.8. Let $(X, \mathcal{O})$ be compact and Hausdorff. Let $A, B \subseteq X$ be closed. Then A and B can be separated from each other by disjoint open neighborhoods.

Exercise 6.9. Every compact subset of a Hausdorff space is closed.

Exercise 6.10. Every continuous function from a compact space into a Hausdorff space is closed.

Exercise 6.11. *Every continuous injective function from a compact space into a Hausdorff space is an embedding.*

Definition 6.12. We call a topological space X *Lindelöf* if every open cover of X has a countable subcover.

Theorem 6.13. Every second-countable space is Lindelöf.

Definition 6.14. We call a topological space X *countably compact* if every countable open cover of X has a finite subcover.

Exercise 6.15. *A countably compact space is compact if and only if it is Lindelöf.*

Theorem 6.16. A metric space is Lindelöf if and only if it is second countable.

Theorem 6.17. Every sequentially compact space is countably compact.

Theorem 6.18. A first-countable space is countably compact if and only if it is sequentially compact.

Theorem 6.19. In a second countable Hausdorff space, compactness, countable compactness and sequential compactness are all equivalent to each other.

Chapter 7

Filters

7.1 Filter Convergence

It would be nice to find a generalization of sequence convergence to non-first-countable spaces that shares many of its useful properties.... It turns out that we can! The correct objects for the task are *filters*.

Exercise 7.1. Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if every convergent filter on X has a unique limit.

Theorem 7.2. Let X be a topological space. Let $U \subseteq X$. Then U is open iff. U is contained in every ultrafilter converging to some point in U .

Proof. $\Rightarrow$: By definition, if U is an open neighborhood of x , any filter converging to x must contain U .

$\Leftarrow$: Let $U \subseteq X$ and assume $U \in \mathcal{U}$ for every ultrafilter converging to a point $u \in U$. Assume U is not open. Then there exists a point $u \in U$ such that every open neighborhood of u contains at least one point in $X \setminus U$. But then any two members of the family

$$\{X \setminus U\} \cup \{U(u)\}$$

have nonempty intersection. Therefore, this family can be extended to an Ultrafilter $\mathcal{U}_u$. Since $\mathcal{U}_u$ contains all open neighborhoods of u by construction, it converges to u , and must thus contain U . Since it also contains $X \setminus U$, we arrive at a contradiction.

□

7.2 Tychonoff's Theorem

Definition 7.3. Let $\mathcal{F}$ be a filter on X . Let $f : X \rightarrow Y$. Then the *pushforward Filter of f on $\mathcal{F}$* is the set

$$f_*(\mathcal{F}) := \{A \subseteq Y \mid p_i^{-1}[A] \in \mathcal{F}\},$$

Exercise 7.4. Verify that the pushforward of a filter is a filter, and that the pushforward of an ultrafilter is an ultrafilter.

Intuitively, the pushforward of a filter under a function generalizes the sequence $f(x_n)$ for some sequence x_n and function f . By extension, the following theorem generalizes the fact that in metric spaces, continuity and sequential continuity are equivalent:

Theorem 7.5: Characterization of continuity in terms of ultrafilters

Let $f : X \rightarrow Y$. Then f is continuous if and only if $f_*(\mathcal{U})$ converges to $f(x)$ for every ultrafilter $\mathcal{U}$ that converges to $x \in X$.

Proof. 1. Assume f is continuous. Let $\mathcal{U}$ be an ultrafilter on X converging to $x \in X$.

Let $U \subseteq Y$ be an open neighbourhood of $f(x)$. Then, since f is continuous, $f^{-1}(U)$ is an open neighbourhood of x . We thus have $f^{-1}(U) \in \mathcal{U}$, which gives us $U \in f_*(\mathcal{U})$ by the definition of the pushforward filter.

Thus, every open neighbourhood of $f(x)$ is contained in $f_*(\mathcal{U})$, so $f_*(\mathcal{U})$ converges to $f(x)$.

2. Assume that the pushforward of any ultrafilter converging to x converges to $f(x)$. Let $U \subseteq Y$ be open.

Let $\mathcal{U}$ be an ultrafilter on X converging to some point $x \in f^{-1}(U)$. Then, by our hypothesis, $f_*(\mathcal{U})$ converges to $f(x)$. Since U is an open neighborhood of $f(x)$, we have $U \in f_*(\mathcal{U})$, and thus $f^{-1}(U) \in \mathcal{U}$. Thus, by theorem 7.2, $f^{-1}(U)$ is open, so f is continuous. $\square$

This theorem generalizes the fact that a sequence in a metric product space converges if and only if every individual component converges.

Theorem 7.6. Let $(X_i, \mathcal{O}_i)_{i \in I}$ be a family of topological spaces. Let $\mathcal{F}$ be a filter on $\prod_{i \in I} X_i$. Then $\mathcal{F}$ converges to a point $x \in \prod_{i \in I} X_i$ if and only if each pushforward filter $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$.

Proof. $\Rightarrow$: Assume that $\mathcal{F}$ converges to $x \in \prod_{i \in I} X_i$. Since all projections p_i are continuous, each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$ by theorem 7.5.

$\Leftarrow$: Assume that each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$. By definition of the product topology, the sets $S_i := p_i^{-1}(U_i)$ for open $U_i \in \mathcal{O}_i$ form a subbase of $\prod_{i \in I} X_i$. Therefore, any open neighborhood $U \in \mathcal{U}(x)$ contains an open neighborhood $B = \bigcap_{i \in J} p_i^{-1}(U_i)$ for some open neighborhoods $U_i \in \mathcal{O}_i$.

Since each $(p_i)_*(\mathcal{F})$ converges to x_i , we have $U_i \in (p_i)_*(\mathcal{F})$ for each i . Therefore, by definition of the pushforward filter, we have $p_i^{-1}(U_i) \in \mathcal{F}$ for all i . Since filters are closed under intersection, we get $B \in \mathcal{F}$,

and since $B \subseteq U$ we also get $U \in \mathcal{F}$ as desired.

□

Theorem 7.7: Tychonoff's Theorem

Let $(X_i, \mathcal{O}_i)_{i \in I}$ be a family of topological spaces. Then if all X_i are compact, $\prod_{i \in I} X_i$ is compact.

Proof. 1. If there exists an $i \in I$ such that $X_i = \emptyset$, then $\prod X_i = \emptyset$ is compact.

2. Now, assume $\prod_{i \in I} X_i \neq \emptyset$. Then there exists an ultrafilter $\mathcal{U}$ on $\prod_{i \in I} X_i$.

Since each X_i is compact, each pushforward $(p_i)_*(\mathcal{U})$ converges to an $x_i \in X_i$. Thus, by theorem 7.6, $\mathcal{U}$ converges to $(x_i)_{i \in I}$ and $\prod_{i \in I} X_i$ is compact.

□

Chapter 8

Higher Separation Properties and Metrizability

8.1 Regular Spaces

Definition 8.1: Regular Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ **regular** if every point can be housed orf from any closed set not containing it:

$$\begin{aligned} \forall X \setminus C \in \mathcal{O} : \\ \forall x \notin C : \\ \exists U_x \in \mathcal{U}(x), U_C \in \mathcal{U}(C) : \\ U_x \cap U_C = \emptyset \end{aligned}$$

We call $(X, \mathcal{O})$ T_3 or **regular Hausdorff**, if it is regular and Hausdorff.

Exercise 8.2. Find a topological space that is regular, but not T_0 .

Exercise 8.3. Show that a topological space is regular Hausdorff (T_3) if and only if it is regular and T_1 .

Definition 8.4: Completely Regular Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ **completely regular** if every point can be separated from any closed set not containing it by a continuous function:

$$\begin{aligned} \forall (X \setminus C) \in \mathcal{O} : \\ \forall x \notin C : \\ \exists f \in C(X, [0, 1]_{\mathbb{R}}) : \\ f(x) = 1, f[A] \subseteq \{0\} \end{aligned}$$

We call $(X, \mathcal{O})$ T_{3a} , $T_{3\frac{1}{2}}$, or **completely regular Hausdorff**, if it is completely regular and Hausdorff.

Exercise 8.5. Find a space that is completely regular, but not T_0 .

Exercise 8.6. Show that a space is completely regular Hausdorff if and only if it is completely regular and T_1 .

Theorem 8.7. Every zero-dimensional space $(X, \mathcal{O})$ is completely regular.

Proof. Since X is zero-dimensional, it has a basis consisting of clopen sets. Given any clopen set U , the function

$$f_U : x \mapsto \begin{cases} 1 & x \in U \\ 0 & x \in X \setminus U \end{cases}$$

is continuous. Now, let C be a closed set not containing x . Then $X \setminus C$ is an open neighbourhood of x . We can now choose a basic (i.e. clopen) neighbourhood $U \subseteq X \setminus C$ of x , at which point f_U is our desired function separating x and C . $\square$

Theorem 8.8. Let $(X, \mathcal{O})$ be completely regular. Then the set

$$\{f^{-1}[U] \mid f \in C((X, \mathcal{O}), [0, 1]_{\mathbb{R}}), U \in \mathcal{O}_{\mathbb{R}}\}$$

is a basis of X .

Proof. 1. Since the f are continuous, every basis set is open.
2. Let $x \in X$. Let U be an open neighbourhood of x . Since X is completely regular, there exists a continuous function f separating $\{x\}$ and $X \setminus U$. Then the basic set $f^{-1}[(0, 1]]$ must be a subset of U , since every element outside of U gets mapped to 0. Thus, every open neighbourhood contains a basic neighbourhood. $\square$

Theorem 8.9: Completely regular cubes

Every completely regular Hausdorff space $(X, \mathcal{O})$ can be embedded into the cube

$$\prod_{f \in C(X, [0, 1])} [0, 1]$$

Proof. We choose the trivial embedding $\iota : x \mapsto (f(x))_{f \in C(X, [0, 1])}$.

1. ι is continuous, since the projections of ι are exactly the elements of $C(X, [0, 1])$, i.e. continuous.
2. Let $x \neq y$. By T_1 , all singletons are closed. Therefore, by complete regularity, there exists a function f with $f(x) = 1$ and $f(y) = 0$. Therefore, we have $\iota(x)_f \neq \iota(y)_f$, i.e. $\iota(x) \neq \iota(y)$.

3. By 8.8, we only need to show openness for the basis sets

$$\{f^{-1}[U] \mid f \in C((X, \mathcal{O}), [0, 1]_{\mathbb{R}}), U \in \mathcal{O}_{\mathbb{R}}\}.$$

For these sets, we get:

$$\begin{aligned} \iota[f^{-1}[U]] &= \iota[(p_f \circ \iota)^{-1}[U]] \\ &= \iota[\iota^{-1}(p_f^{-1}[U])] \\ &= \iota \left[\iota^{-1} \left[U \times \prod_{g \neq f} [0, 1] \right] \right] \\ &= \left(U \times \prod_{g \neq f} [0, 1] \right) \cap \iota[X] \end{aligned}$$

Where $\left(U \times \prod_{g \neq f} [0, 1] \right)$ is open, since every component besides the f -component is open.

□

8.2 Normal Spaces

Definition 8.10: Normal Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ **normal** if every pair of disjoint closed sets can be housed orf from each other by disjoint neighbourhoods:

$$\begin{aligned} &\forall C_1, C_2 \in \mathcal{C} : \\ &C_1 \cap C_2 = \emptyset \implies \\ &\exists U_1 \in \mathcal{U}(C_1), U_2 \in \mathcal{U}(C_2) : \\ &U_1 \cap U_2 = \emptyset \end{aligned}$$

We call $(X, \mathcal{O})$ **T_4** , or **normal Hausdorff**, if it is normal and Hausdorff.

Exercise 8.11. Find a topological space that is normal, but not T_0 .

Lemma 8.12: Urysohn's Lemma

A topological space is normal if and only if any two disjoint closed subsets can be separated by a continuous function.

Corollary 8.13. Every normal Hausdorff space is regular Hausdorff.

8.3 Urysohn's metrization theorem

Chapter 9

Lattice Theory

9.1 Lattices

Definition 9.1. A partially ordered set $(L, \leq)$ is called a *lattice* if every pair of elements $x, y \in L$ has a least upper bound $x \vee y$, also known as a *join*, and a greatest lower bound $x \wedge y$, also known as a *meet*. A *complete lattice* is a lattice in which every subset $A \subseteq L$ has a join $\sup A$ and a meet $\inf A$.

Definition 9.2. Given a complete lattice L , the *bottom element* is the element $0 := \inf L$, and the *top element* is the element $\infty := \sup L$.

Definition 9.3. Let L be a complete lattice. Let $x, y \in L$. Then we say that x is *well above* y , denoted $x > y$, iff for any subset $A \subseteq L$,

$$y \geq \inf A \implies \exists a \in A : x \geq a$$

Lemma 9.4. Let L be a complete lattice. Let $x, y, z \in L$. Then:

1. $x > y$ implies $x \geq y$,
2. $x > y$ and $y \geq z$ implies $x \geq z$,
3. $x \geq y$ and $y > z$ implies $x > z$.

Proof. 1. We have $y \geq \inf \{y\}$, so $x > y$ implies $x \geq y$.

2. Follows from applying transitivity to the first part of the lemma.

3. $z \geq \inf A \implies \exists a \in A : x \geq y \geq a$.

□

Lemma 9.5. Let L be a complete lattice. Let $x \in L$ and $A \subseteq L$. Then

$$x > \inf A \Leftrightarrow \exists a \in A : x > a$$

9.2 Value Quantales

9.3 Generalized metrics metrize every topological space

Part V

Algebraic Topology

Chapter 1

Homotopy

Definition 1.1

Let X, Y be topological spaces. Let $f, g : X \rightarrow Y$ be continuous. Then f is called **homotopic** to g via a map h iff. h is a "continuous interpolation between f and g , i.e:

1. h is a continuous map $X \times [0, 1] \rightarrow Y$,
2. For all $x \in X$, we have $h(x, 0) = f(x)$,
3. For all $x \in X$, we have $h(x, 1) = g(x)$.

We call h a **free homotopy**, or simply a **homotopy**, between f and g . We write $f \sim_h g$ if f is a homotopy between f and g . We write $f \sim g$ if there exists a homotopy between f and g .

Exercise 1.2. Show that the property of being homotopic is an equivalence relation.

Definition 1.3. We say that f and g are **homotopic relative to a subset $A \subseteq X$** and write $f \sim_{\text{rel } A} g$ or $f \sim_{h, \text{rel } A} g$ iff:

1. $f \sim_h g$,
2. For all $t \in [0, 1]$, we have $h(t, \cdot)|_A = f|_A = g|_A$

I personally find the designation "homotopic relative to A " quite unintuitive, given that the property of being homotopic relative to a given subset is stronger than the property of being homotopic.

Exercise 1.4. Show that $f \sim_h g$ is equivalent to $f \sim_{h, \text{rel } \emptyset} g$.

Exercise 1.5. Let $f_0, f_1 : X \rightarrow Y$ and $g_0, g_1 : Y \rightarrow Z$ be continuous. Let h_0 be a homotopy between f_0 and f_1 and let h_1 be a homotopy between f_1 and g_1 . Then $H : (x, t) \mapsto h_1(h_0(x, t), t)$ is a homotopy between $g_0 \circ f_0$ and $g_1 \circ f_1$.

Definition 1.6

Let $f : X \rightarrow Y$ be continuous. Then f is called a **homotopy equivalence** between X and Y iff. there exists a map $g : Y \rightarrow X$ such that $g \circ f$ is homotopic to id_X and $f \circ g$ is homotopic to id_Y . We write $X \simeq_{f, g} Y$, $X \simeq_f Y$, or simply $X \simeq Y$.

Exercise 1.7. Show that the property of being homotopy equivalent is an equivalence relation.

Exercise 1.8. Show that homeomorphism implies homotopy equivalence.

Definition 1.9. A topological space X is called *contractible* iff. X is homotopy equivalent to a one-point space.

Exercise 1.10. 1. Show that

$$h : (x, t) \mapsto x - tx$$

is a homotopy between $\text{id}_{\mathbb{R}^n}$ and $f : \mathbb{R}^n \mapsto \{0\}$.

2. Conclude that $\mathbb{R}^n$ is contractible.

3. Conclude that homotopy equivalence does not imply homeomorphism.

Exercise 1.11. A topological space X is contractible if and only if there exists a point $x_0 \in X$ such that id_X is homotopic to the constant map $X \rightarrow x_0$.

Definition 1.12. Let X be a subset of a topological vector space. Then we call X *star-shaped* iff. there exists a point $x_0 \in X$ such that X contains the connecting line between x_0 and any other point $x \in X$, i.e.

$$\begin{aligned} \forall x \in X : \\ \forall t \in [0, 1] : \\ (1 - t)x_0 + tx \in X \end{aligned}$$

Exercise 1.13. Show that any star-shaped set is contractible. Conclude that any convex set is contractible.

Lemma 1.14. Let X be a topological space. Let $A \subseteq X$. Then A and X are homotopy equivalent if there exists a homotopy between id_X and $\text{id}_X|_A$, i.e. a continuous map $h : X \times [0, 1] \rightarrow X$ such that for all $x \in X$:

1. $h(x, 0) = x$,
2. $h(x, 1) \in A$,
3. $x \in A \implies h(x, 1) = x$

1.1 The Fundamental Group

Definition 1.15. Let X be a topological space. Then a continuous map $\gamma[0, 1] \rightarrow X$ with $\gamma(0) = \gamma(1)$ is known as a *loop*. We denote the set of all loops in X with $\gamma(0) = \gamma(1) = x_0$ as $\Omega(X, x_0)$.

Lemma 1.16. Let $\gamma : [0, 1] \rightarrow X$ be a loop. Let $\varphi : [0, 1] \rightarrow [0, 1]$ be a continuous map such that $\varphi(0) = 0$ and $\varphi(1) = 1$. Then $\gamma \circ \varphi$ and γ are homotopic relative to $\{0, 1\}$.

Proof. The map

$$h : (x, t) \mapsto \gamma((1-t)x + t\varphi(x))$$

is a homotopy between γ and $\gamma \circ \varphi$. □

Definition 1.17. Let $f, g \in \Omega(X, x_0)$. Then we define the *concatenation of f and g* as the map

$$(f * g)(t) = \begin{cases} f(2t) & 0 \leq t \leq 1/2, \\ g(2t - 1) & 1/2 \leq t \leq 1. \end{cases}$$

Definition 1.18. Let $\gamma \in \Omega(X, x_0)$. We define the *inverse loop* of γ to be the loop that “moves through gamma backwards”, i.e.

$$\bar{\gamma}(t) := \gamma(1 - t)$$

Exercise 1.19. Verify that $\bar{\bar{\gamma}} = \gamma$.

Exercise 1.20. Let $\gamma \in \Omega(X, x_0)$. Then $\gamma * \bar{\gamma}$ is homotopic to $e_{x_0} : X \rightarrow \{x_0\}$ relative to $\{0, 1\}$.

Theorem 1.21: Existence of the fundamental group

Let X be a topological space. Let $x_0 \in X$. Then the quotient space

$$\pi_1(X, x_0) = \Omega(X, x_0) / \sim_{\text{rel } (0,1)}$$

of loops modulo homotopy becomes a group, known as the *fundamental group of (X, x_0)* , when equipped path concatenation

$$[f] * [g] = [f * g],$$

as the group operation. The neutral element of this group is $e_{x_0} := (x \mapsto x_0)$.

Theorem 1.22. If we additionally define

$$\begin{aligned} \pi_1(f) : \pi_1(X, x) &\mapsto \pi_1(Y, f(x)) \\ [\gamma] &\mapsto [f \circ \gamma] \end{aligned}$$

for all continuous $f : X \rightarrow Y$, the map π_1 becomes a covariant functor from the category of topological spaces into the category of groups.

Definition 1.23. Whenever we need to make the base point clear, we write $\pi_1(f, x) := \pi_1(f) : \pi_1(X, x) \mapsto \pi_1(Y, f(x))$

Theorem 1.24: Fundamental groups at different base points

Let X be a topological space. Let $x, y \in X$. Then every continuous path $\delta : [0, 1] \rightarrow X$ induces a group isomorphism

$$J_\delta : \pi_1(X, \delta(0)) \cong \pi_1(X, \delta(1))$$

$$[\gamma] \mapsto [\bar{\delta} * \gamma * \delta]$$

Proof. 1. Since loop concatenation preserves homotopy, J_δ sends different representatives of the same equivalence class to the same equivalence class.

2. J_δ goes from $\pi_1(X, x)$ to $\pi_1(X, y)$, since for any $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = \gamma(1) = x$, $\bar{\delta}$ traces out a path from $\delta(1)$ to $\delta(0)$, then γ traces out a loop from $\delta(0)$ to $\delta(0)$, and then δ traces out a path from $\delta(0)$ back to $\delta(1)$, i.e. $\bar{\delta} * \gamma * \delta$ is a loop from $\delta(1)$ to $\delta(1)$.

3. J_δ is a group homomorphism, since

$$\begin{aligned} J_\delta([\gamma_1 * \gamma_2]) &= [\bar{\delta} * \gamma_1 * \gamma_2 * \delta] \\ &= [\bar{\delta} * \gamma_1 * e_{x_0} * \gamma_2 * \delta] \\ &= [\bar{\delta} * \gamma_1 * \delta * \bar{\delta} * \gamma_2 * \delta] \\ &= [\bar{\delta} * \gamma_1 * \delta] * [\bar{\delta} * \gamma_2 * \delta] \\ &= J_\delta([\gamma_1]) * J_\delta([\gamma_2]) \end{aligned}$$

(Recall that any map that respects the group operation also respects the neutral element and inverses, and is thus a homomorphism.)

4. J_δ is bijective with $J_\delta^{-1} = J_{\bar{\delta}}$:

$$\begin{aligned} J_\delta(J_{\bar{\delta}}(\gamma)) &= [\bar{\delta} * (\delta * \gamma * \bar{\delta}) * \delta] = [\gamma] \\ J_{\bar{\delta}}(J_\delta(\gamma)) &= [\delta * (\bar{\delta} * \gamma * \delta) * \bar{\delta}] = [\gamma] \end{aligned}$$

□

Corollary 1.25. If X is path connected, $\pi_1(X, x) \cong \pi_1(X, y)$ for any $x, y \in X$.

Definition 1.26. Let $\gamma_0 \in \Omega(X, x)$ and $\gamma_1 \in \Omega(X, y)$. Then we call $h : [0, 1] \times [0, 1] \rightarrow X$ a **free homotopy of loops**, or simply **loop homotopy**, between γ_0 and γ_1 iff. h is a homotopy between γ_0 and γ_1 such that $h_t(x) := h(x, t)$ is a loop for any fixed $t \in [0, 1]$.

Exercise 1.27. 1. Verify that h is a loop homotopy iff. h is a homotopy and $h_t(0) = h_t(1)$ for any $t \in [0, 1]$.

2. Show that any $\{0, 1\}$ -homotopy is a loop homotopy.

The additional constraint that $h(\cdot, t)$ is a loop may seem superfluous, but there are in fact homotopies between loops that aren't loop homotopies:

- Exercise 1.28.**
1. Let $\gamma_0 = \gamma_1 : [0, 1] \rightarrow \{0\}$ be constant loops. Let $h(x, t) := xt(1-t)$. Then h is a homotopy between γ_0 and γ_1 , but fails to be a loop homotopy.
 2. Let $\gamma_0(x) = e^{2\pi ix}$. Let $\gamma_1(x) = e^{4\pi ix}$. Then $h(x, t) := e^{2\pi i(1+t)x}$ is a homotopy between γ_0 and γ_1 , but fails to be a loop homotopy.
 3. Assume that any loop $\gamma : [0, 1] \rightarrow S^1$ with 1 as its base point is homotopic a loop of the form $\gamma(x) := e^{2\pi inx}$ for some $n \in \mathbb{Z}$ (we will prove this later). Does 2. imply that the fundamental group $\pi_1(S^1, 1)$ is trivial?

Definition 1.29. A loop $\gamma \in \Omega(X, x)$ is called **contractible in X** iff. there exists an $x \in X$ and a loop homotopy h such that $\gamma \sim_h e_x$.

Lemma 1.30. $\gamma \in \Omega(X, x)$ is contractible in X if $\gamma \sim_{\text{rel } 0,1} e_{\gamma(0)}$.

Theorem 1.31. Let $\gamma \in \Omega(X, x)$ be a loop. Let $S^1 \cong I/\{0, 1\}$ be equipped with the quotient topology. Then γ is contractible if and only if $\gamma \circ p : S^1 \rightarrow X$ has a continuous extension to the closed unit ball $\bar{B}^2$.

Theorem 1.32. Let h be a homotopy between $f, g : X \rightarrow Y$. Let $x_1 \in X$. Then

$$\begin{aligned} h(x_1, \cdot) : [0, 1] &\rightarrow Y \\ t &\mapsto h(x_1, t) \end{aligned}$$

is a path from $f(x_1)$ to $g(x_1)$ which fulfills

$$\pi_1(g, x_1) = J_{h(x_1, \cdot)} \circ \pi_1(f, x_1)$$

Recall that $\pi_1(g, x_1)$ is the map $[\gamma] \mapsto [g \circ \gamma]$, and that J_δ is the map $[\gamma] \mapsto [\bar{\delta} * \gamma * \delta]$, so this theorem states that

$$[g \circ \gamma] = [\bar{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)]$$

for any $\gamma \in \pi_1(X, x_1)$, or, equivalently, that the diagram

$$\begin{array}{ccc} & \pi_1(X, x_1) & \\ \swarrow \pi_1(f, x_1) & & \searrow \pi_1(g, x_1) \\ \pi_1(Y, f(x_1)) & \xrightarrow{J_{h(x_1, \cdot)}} & \pi_1(Y, g(x_1)) \end{array}$$

commutes.

Proof. Define

$$H_t(x) := h(x_1, t + (1-t)x).$$

We have $(f \circ \gamma) \in \Omega(X, f(x_1))$ and $(g \circ \gamma) \in \Omega(X, g(x_1))$, since

$$f(\gamma(0)) = f(x_1) = f(\gamma(1))$$

$$g(\gamma(0)) = g(x_1) = g(\gamma(1))$$

Then a homotopy between $g \circ \gamma$ and $\bar{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)$ is given by:

$$k(x, t) := (\bar{H}_t * (h(\cdot, t) \circ \gamma) * H_t)(x),$$

since we get:

$$\begin{aligned} k(x, 0) &= (\bar{H}_0 * (h(\cdot, 0) \circ \gamma) * H_0)(x) \\ &= (\bar{H}_0 * (f \circ \gamma) * H_0)(x) \\ &= (\bar{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot))(x) \end{aligned}$$

and

$$\begin{aligned} k(x, 1) &= (\bar{H}_1 * (h(\cdot, 1) \circ \gamma) * H_1)(x) \\ &= (h(x_1, (1-1) + (1-(1-1))(\cdot)) * (g \circ \gamma) * h(x_1, 1 + (1-1)(\cdot)))(x) \\ &= (h(x_1, \cdot) * (g \circ \gamma) * h(x_1, 1))(x) \\ &= (e_{g(x_1)} * (g \circ \gamma) * e_{g(x_1)})(x) \\ &= (g \circ \gamma)(x), \end{aligned}$$

where the second to last step follows since the loop $(g \circ \gamma)$ ends on the base point $g(x_1)$. $\square$

Lemma 1.33. Let $\gamma : [0, 1] \rightarrow X$ be continuous. Let $f : X \rightarrow Y$. Then

$$\pi_1(f, \gamma(1)) \circ J_\gamma = J_{f \circ \gamma} \circ \pi_1(f, \gamma(0)),$$

i.e. the diagram

$$\begin{array}{ccc} \pi_1(X, \gamma(0)) & \xrightarrow{\pi_1(f, \gamma(0))} & \pi_1(Y, f(\gamma(0))) \\ \downarrow J_\gamma & & \downarrow J_{f \circ \gamma} \\ \pi_1(X, \gamma(1)) & \xrightarrow{\pi_1(f, \gamma(1))} & \pi_1(Y, f(\gamma(1))) \end{array}$$

commutes.

Proof. Let $\sigma \in \Omega(X, \gamma(0))$. Then we have

$$\begin{aligned} \pi_1(f, \gamma(1)) \circ J_\gamma([\sigma]) &= \pi_1(f, \gamma(1))([\bar{\gamma} * \sigma * \gamma]) \\ &= [f \circ (\bar{\gamma} * \sigma * \gamma)] \\ &= [(f \circ \bar{\gamma}) * (f \circ \sigma) * (f \circ \gamma)] \\ &= [\overline{(f \circ \gamma)} * (f \circ \sigma) * (f \circ \gamma)] \\ &= J_{f \circ \gamma}([f \circ \sigma]) \\ &= J_{f \circ \gamma} \circ \pi_1(f, \gamma(0))([\sigma]) \end{aligned}$$

$\square$

Theorem 1.34. Let $f : X \rightarrow Y$ be a homotopy equivalence. Let $x_0 \in X$. Then the map $\pi_1(f, x_0)$ is a group isomorphism $\pi_1(X, x_0) \cong \pi_1(Y, f(x_0))$.

Chapter 2

Fibrations and Coverings

Definition 2.1: Locally Trivial Fibration

Let $\pi : Y \rightarrow X$ be continuous. Then π is called a *locally trivial fibration* iff. for every $x \in X$, there exists an open neighborhood U such that there exists a homeomorphism $h_U : \pi^{-1}[U] \cong U \times \pi^{-1}[x]$ fulfilling $p_1 \circ h_U = \pi|_{\pi^{-1}[U]}$.

$\pi^{-1}[x]$ is called the **Fiber of π at x** , and sometimes written F_x .

$$\begin{array}{ccc} Y \supseteq \pi^{-1}[U] & \xrightarrow{h_U} & U \times \pi^{-1}[x] \\ \downarrow \pi & & \swarrow p_1 \\ X \supseteq U & & \end{array}$$

Definition 2.2. Let $\pi : Y \rightarrow X$ be a locally trivial fibration. Then:

1. If all $\pi^{-1}[x]$ are homeomorphic, we call π a **fiber bundle**.
2. If every $\pi^{-1}[x]$ is discrete, we call π a **covering map**.
3. If we can pick $U = X$, π is called a **trivial fibration**.

Exercise 2.3. If we replace $\pi^{-1}[x]$ in the definition by an arbitrary topological space F_U , we get $F_U \cong \pi^{-1}[x]$. Thus, both definitions are equivalent.

Lemma 2.4. Let $\pi : Y \rightarrow X$ be a locally trivial fibration. Let X be connected. Then π is a trivial fibration.

Proof. Let $x_0 \in X$. Let $V = \{x \in X \mid F_x \cong F_{x_0}\}$ be the set of all points whose fibers are homeomorphic to the fiber of x_0 . Let U be a neighborhood as in the definition. Then the fibers of all $x \in U$ must be homeomorphic by definition. Thus, every point in V has a neighborhood contained in V , so V is open. We have

$$X \setminus V = \bigcup_{y \notin V} \{x \in X \mid F_x \cong F_y\} := \bigcup_{y \notin V} V_y$$

Since every V_y is open by the same argument, $X \setminus V$ must be open. But since X is connected with $(X \setminus V) \cup V = X$ and V nonempty, $X \setminus V$ must

be empty. □

Theorem 2.5. A continuous function $\pi : Y \rightarrow X$ is a covering map if and only if every $x \in X$ has an open neighborhood U such that $\pi^{-1}[U]$ can be decomposed into disjoint open sets $V_i \subseteq Y$ such that $\pi|_{V_i}$ is a homeomorphism $V_i \cong U$.

Proof. $\Rightarrow$: Let π be a covering map. Let $I = F_U$. Since h_U is bijective, the sets $V_i := h_U^{-1}[U \times \{i\}]$ are disjoint. Since h_U is a homeomorphism and $I = F_U$ is discrete, the V_i are also open. Since $p_1|_{U \times \{i\}}$ is a homeomorphism $U \times \{i\} \cong U$, the composition

$$\pi|_{V_i} = (p_1|_{U \times \{i\}}) \circ h|_{V_i}$$

must be a homeomorphism $V_i \mapsto U$.

$\Leftarrow$: Define

$$\begin{aligned} h_U : \pi^{-1}[U] &\cong U \times I \\ y \in V_i &\mapsto (\pi(y), i). \end{aligned}$$

Then the restriction of h_U to each V_i is by definition a homeomorphism $V_i \cong U \cong U \times \{i\}$, and so the whole of h_U is a homeomorphism $\pi^{-1}[U] = \bigsqcup_{i \in I} V_i \cong \bigsqcup_{i \in I} U \times \{i\} = U \times I$. □

Definition 2.6: Lifting

Let $\pi : Y \rightarrow X$ be a locally trivial fibration. Let $f : Z \rightarrow X$ be continuous. Then a continuous map $\hat{f} : Z \rightarrow Y$ is called a **lift from f to Y** if $\pi \circ \hat{f} = f$.

Any path can be lifted:

Theorem 2.7. Let $\pi : Y \rightarrow X$ be a locally trivial fibration. Let $\gamma : [0, 1] \rightarrow X$ be continuous. Let $\pi(y_0) = \gamma(0)$. Then there exists a lift $\hat{\gamma} : [0, 1] \rightarrow Y$ with $\hat{\gamma}(0) = y_0$ such that $\pi \circ \hat{\gamma} = \gamma$.

And lifts respect fundamental groups:

Theorem 2.8. Let $\pi : Y \rightarrow X$ be a locally trivial fibration. Let $f : Z \rightarrow X$. Let $\hat{f} : Z \rightarrow Y$ be a lift of f . Then we have:

$$\begin{aligned} \forall z_0 \in Z : \\ \exists y_0 \in Y : \\ \hat{f}(z_0) = y_0, \\ \pi_1(f)[\pi_1(Z, z_0)] \subseteq \pi_1(\pi)[\pi_1(Y, y_0)] \end{aligned}$$

Proof.

$$\begin{aligned}\pi_1(f)[\pi_1(Z, z_0)] &= (\pi_1(\pi \circ \hat{f}))[\pi_1(Z, z_0)] \\ &= (\pi_1(\pi) \circ \pi_1(\hat{f}))[\pi_1(Z, z_0)] \\ &= \end{aligned}$$

□